



The effect of drying, cell disruption and storage on the sensory properties of *Nannochloropsis* sp.

Bert Coleman^{a,b,*}, Christof Van Poucke^c, Bavo De Witte^a, Valentina Casciaro^a,
Tanja Moerdijk-Poortvliet^d, Koenraad Muylaert^b, Johan Robbens^a

^a Aquatic Environment and Quality, Cell Blue Biotech and Food Integrity, Flanders Research Institute for Agriculture, Fisheries and Food (ILVO), Jacobsenstraat 1, 8400 Ostend, Belgium

^b Biology Department KULAK, KU Leuven Kulak, Etienne Sabbelaan 53, 8500 Kortrijk, Belgium

^c Technology and Food Science Unit, Flanders Research Institute for Agriculture, Fisheries and Food (ILVO), Brusselsesteenweg 370, 9090 Melle, Belgium

^d HZ University of Applied Sciences, Department of Technology, Water and Environment, Research Group Marine Biobased Chemistry, P.O. Box 364, 4380 AJ Vlissingen, the Netherlands

ARTICLE INFO

Keywords:

Microalgae
Processing
Storage
High pressure homogenization
Sensory evaluation
Volatile organic compounds

ABSTRACT

Adding microalgae into food could strongly influence the flavor of the final product which is the key factor for consumer acceptance. To assist food processors in the development of tasteful food products containing microalgae, it is necessary to understand the impact of processing and storage on the flavor of freshly harvested microalgae biomass. In this study, the effect of commonly used drying techniques (freeze-drying, spray drying and thin-layer oven drying) and cell disruption (high pressure homogenization) on the sensory quality of fresh *Nannochloropsis* sp. biomass was investigated. Furthermore, the impact of storage on the flavor properties of intact (non-disrupted) and disrupted *Nannochloropsis* paste was examined. Sensory evaluation by a trained panel indicated that the flavor of *Nannochloropsis* was not affected by freeze-drying. Whereas thin-layer oven drying at 60 °C resulted in a strong bitter taste and a musty odor. This odor is positively correlated with volatile organic compounds (VOCs) including Strecker aldehydes that arise from the Strecker degradation in the course of the Maillard reaction. Spray drying reduced the odor intensity and grassy odor of *Nannochloropsis* which could be attributed to the loss of several VOCs including esters, saturated alcohols and dimethyl sulfide. Furthermore, cell disruption of *Nannochloropsis* using high pressure homogenization resulted in intense grassy and fish oil odors features due to high amounts of fatty acid-derived unsaturated aldehydes, ketones and alcohols. Finally, dark storage of fresh intact and disrupted *Nannochloropsis* paste at ambient temperature strongly impacts the volatile profile due to the rapid formation of several VOCs, which negatively impact the palatability of the *Nannochloropsis* biomass. Our results provide the basis to be applied in downstream food processing of harvested *Nannochloropsis* biomass in order to maintain or optimize the flavor properties for specific food products.

1. Introduction

Marine microalgae are increasingly being explored as a sustainable and nutritious source of biomass. The cultivation of these microalgae do not require arable land and freshwater, avoiding competition for the resources with feed and food production. The microalgae biomass can be high in protein content and could possess polyunsaturated fatty acids (PUFAs), vitamins and minerals [1]. Some marine microalgae species have an interesting flavor profile that closely matches the flavor of seafood [2,3]. Microalgae produce volatile organic compounds (VOCs)

such as unsaturated aldehydes, ketones and alcohols, as well as sulfur- and nitrogen-containing compounds that are found in many seafood products [4–8]. Additionally, some microalgae are rich in free amino acids (FAAs) and 5'-nucleotides producing a strong umami taste, which is a characteristic feature of most seafood [7,9,10]. Marine microalgae are already used for their flavor profile in contemporary cuisine [11]. The strong seafood flavor of microalgae opens up future possibilities to use microalgae as flavoring agents in plant-based seafood products (e.g. vegan fish and shellfish). As flavor is the determining factor for consumer acceptance [12], characterizing the flavor of edible microalgae is

* Corresponding author at: Aquatic Environment and Quality, Cell Blue Biotech and Food Integrity, Flanders Research Institute for Agriculture, Fisheries and Food (ILVO), Jacobsenstraat 1, 8400 Ostend, Belgium.

E-mail address: bert.coleman@ilvo.vlaanderen.be (B. Coleman).

<https://doi.org/10.1016/j.algal.2023.103092>

Received 16 December 2022; Received in revised form 3 March 2023; Accepted 3 April 2023

Available online 5 April 2023

2211-9264/© 2023 Elsevier B.V. All rights reserved.

essential to develop delicious food products containing algae and to be able to fully explore the culinary and flavoring aspect of microalgae.

Microalgae are cultured as a dilute suspension in an aqueous culture medium. During harvesting, the microalgae cells are concentrated, which is in most commercial facilities done by centrifugation. This produces a wet paste with a dry weight (DW) content of about 10–20 %. Because some microalgae cells are surrounded by a rigid cell wall, cell disruption techniques are often applied to increase the digestibility of the biomass and the bioaccessibility of nutrients [13]. Furthermore, the freshly harvested microalgae paste is often stored for some time, either prior to further processing or until consumption. For longer term storage, the microalgae paste can either be stored frozen or it can be dried to a powder to reduce the water content. However, it is not known to what extent these different processing methods influence the flavor of the microalgae.

Previous studies have shown that storage, cell disruption, freezing or drying can result in chemical modifications of the microalgae biomass. Balduyck et al. [14] showed that lipolysis of lipids into free fatty acids occur directly after harvesting by centrifugation of *T-Isochrysis lutea* and continues during storage of the wet microalgae paste. [15] observed a reduction of the carbohydrate content, an increase of the microbial load and a rise in the amount of organic acids during the storage of concentrated *Porphyridium purpureum* paste. A commonly used method for cell disruption of microalgae is high pressure homogenization (HPH) [16]. Balduyck et al. [17] showed that HPH-treated *Nannochloropsis* results in an accelerated lipolysis compared to the intact (non-disrupted) microalgae cells, which can most likely be explained by the activation of endogenous lipolytic enzymes. Freezing of microalgae paste promotes the formation of intercellular ice crystals which might disrupt microalgae that lack a distinct cell wall [18]. Drying of microalgae biomass for human consumption is usually achieved by either heating (spray drying, thin-layer oven drying) or applying a vacuum (freeze-drying) [19]. The use of heat during drying can result in the degradation of heat-sensitive compounds, depending on the exposure temperature and time. Ryckebosch et al. [20] observed a 25 % loss of carotenoid content after spray drying of *Phaeodactylum tricornutum* compared to fresh and freeze-dried microalgae biomass. Oliveira et al. [21] observed a loss in phycocyanin content and an increase of lipid oxidation after thin-layer oven drying at 55 °C of *Spirulina platensis* compared to the fresh biomass. Hosseinizand et al. [22] observed a color loss and a reduction of lipid content after thin-layer oven drying of *Chlorella vulgaris* compared to the freeze-dried biomass and recommended a drying temperature between 60 and 80 °C. Freeze-drying avoids heating and is therefore often applied to retain the quality of the final product.

Changes in the biomass composition that occur during storage, after cell disruption, freezing or drying may also influence the flavor properties of microalgae. Research on microalgae in an ecological context have shown that many microalgae species release VOCs in response to aging and cell rupture. These compounds include unsaturated aldehydes such as 2,4-decadienal [23], degradation products of carotenoids such as β -cyclocitral [24] and degradation products of the sulfur-containing compound dimethylsulfoniopropionate (DMSP) such as dimethyl sulfide [25]. These VOCs are highly odor-active because of their low odor threshold values (OTVs) and therefore could affect the flavor of the microalgae. In addition, the PUFAs are prone to enzymatic (lip-oxygenase/lyase) and chemical oxidation (autooxidation) and may result in the formation of different odor-active VOCs. Cell disruption and freezing of microalgae paste could also induce changes in their protein structure, affecting the FAAs content [26,27]. The increase of FAAs may affect the taste intensity of microalgae, as is seen after microbial processing (e.g. fermentation) of many protein-rich foods [28]. Finally, higher temperatures applied during drying could promote thermal degradation of proteins and lipids and give rise to chemical reactions such as the Maillard reaction and Strecker degradation in which amino acids and reducing sugars or carbonyl groups react, resulting in new flavor compounds [29].

The aim of this study is to assess the effect of processing and storage on the flavor properties of *Nannochloropsis* biomass. This marine microalgae was evaluated for taste and aroma using a combination of sensory evaluation by a trained sensory panel and chemical flavor profiling. The latter comprises the analysis of the VOC profile and taste compounds including FAAs and 5'-nucleotides. *Nannochloropsis* was selected because it is an excellent nutritional source of long-chain PUFAs such as eicosapentaenoic acid (EPA, C20:5 n-3). Moreover, this microalgae possesses a rigid cell wall, containing the non-hydrolysable biopolymer algaenan which makes it possible to create a clear difference between disrupted and intact microalgae [30]. *Nannochloropsis* is already used as a food ingredient in food products (e.g. pasta, biscuits and bread) where it strongly affects the consumer acceptance because of its fishy flavor [31,32]. We hypothesize that both processing and storage can affect the aroma as well as the taste of the *Nannochloropsis* biomass. This would provide essential information for the optimal exploitation of *Nannochloropsis* as a food ingredient or flavoring agent.

2. Materials and methods

2.1. Microalgae cultivation

Nannochloropsis sp. (strain CCAP278/11) was cultivated at Proviron Industries nv (Hemiksem, Belgium) as described by Fret et al. [33] with slight modifications. Briefly, *Nannochloropsis* sp. was grown in semi-continuous aerated vertical flat panel photobioreactors (PBRs), using the in-house developed ProviAPT technology (Provion's Advanced Photobioreactor Technology). The PBRs were illuminated by LEDs and provided with artificial seawater and food-grade nutrients. The *Nannochloropsis* culture was concentrated by centrifugation (Evodos, The Netherlands) until a water content of 90.1 ± 0.1 %. This water content allowed to pump the biomass paste during processing (e.g. high pressure homogenization and spray drying). Importantly, as cultivation conditions might also influence the flavor of microalgae [34], the processing and storage tests were performed on a large batch from the same *Nannochloropsis* culture harvested on a single day. Approximately 60 L of the freshly harvested *Nannochloropsis* paste (10 % DW) was acquired and kept refrigerated (4 °C) in the dark for no >24 h until processing (drying and cell disruption).

2.2. Biomass storage and processing

The refrigerated *Nannochloropsis* paste was stirred and divided in batches for further processing as shown in Fig. 1. Part of the biomass (5 L) was stored in the dark without processing for 24 h at ambient temperature (20 °C) (further referred to as 'intact *Nannochloropsis*'). Another part of the biomass (20L) was disrupted using high pressure homogenization (further referred to as 'disrupted *Nannochloropsis*'). The *Nannochloropsis* paste was passed four times through the HPH-system (GEA Panther Model NS3006L) at a pressure of 100 MPa with a feed velocity of 100 L/h according to Bernaerts et al. [16], which resulted in complete disruption of the *Nannochloropsis* cells. To avoid heating, the outlet was cooled through a double jacket tube (4 °C) which allowed to keep the temperature of the paste below 17 °C. During cell disruption, the homogenized product was continuously flushed with nitrogen to minimize (surface) oxidation reactions. Subsequently, the disrupted *Nannochloropsis* paste was stored in the dark for 24 h at ambient temperature. During the dark storage of the fresh intact and disrupted *Nannochloropsis* pastes, several subsamples were taken for volatiles analysis. Furthermore, the subsamples taken immediately after cell disruption (0 h) and after storage (24 h) were frozen at -80 °C and lyophilized for analysis of free amino acids and 5'-nucleotides. In addition, fresh subsamples of intact and disrupted *Nannochloropsis* (0 h) were frozen and stored at -80 °C for volatile analysis and sensory analysis. Subsamples were stored for no longer than 2 months until further analyses.

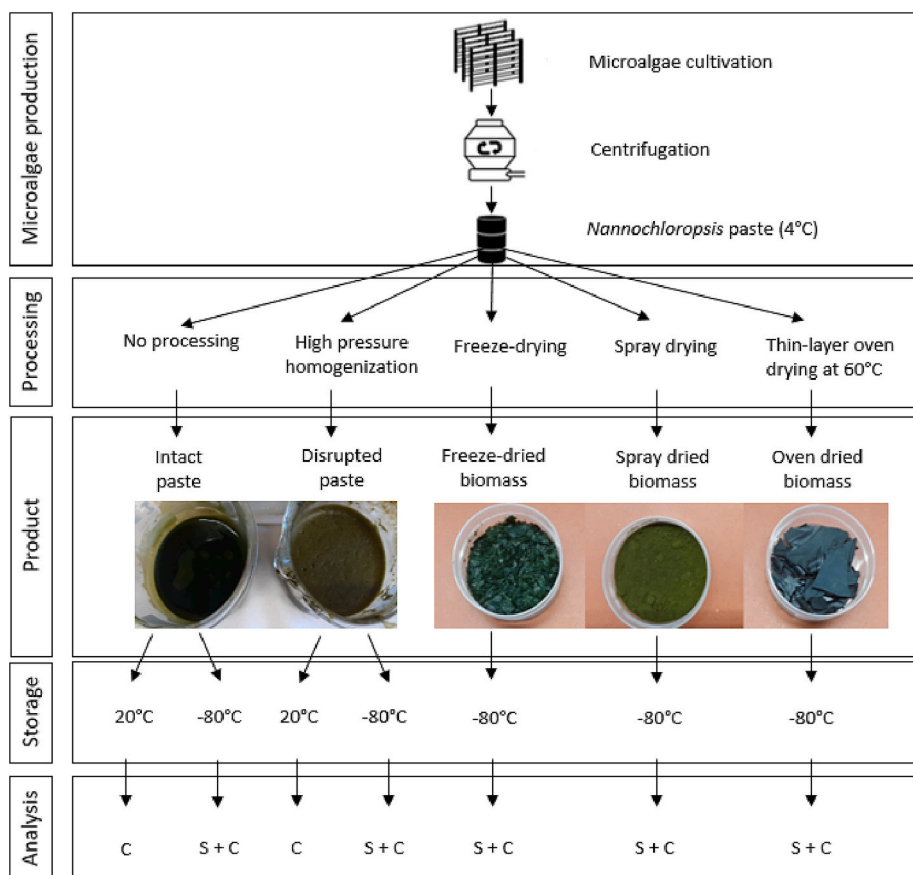


Fig. 1. Schematic overview of the production, processing and storage of *Nannochloropsis* biomass and the analyses performed on the biomass. C stands for chemical flavor analysis and S for sensory evaluation.

The remaining *Nannochloropsis* paste was dried using three different drying methods. Part of the biomass (4 L) was frozen at -25°C and subsequently freeze-dried in a Epsilon 2–10 D LSC (Martin-Christ) freeze-dryer. In the freeze-dryer, the temperature of the microalgae paste was lowered to -50°C and maintained for 2 h at atmospheric pressure. During the sublimation step, the temperature was raised from -50°C to 15°C in 20 h (3.25°C/h) at an absolute pressure of 16 Pa. Afterwards, the temperature of 15°C and the pressure of 16 Pa was held for another 20 h. In the desorption step, the bound water was removed by applying a temperature of 20°C at a pressure of 2.5 Pa for 5 h. Another part of the biomass (30 L) was spray dried using an Anhydro SPX semi-industrial spray-drier at a feed rate of 48.6 L/h. The inlet and outlet air temperatures of the spray drier were set to $140 \pm 5^{\circ}\text{C}$ and $90 \pm 5^{\circ}\text{C}$, respectively. The atomizer wheel rotated at a maximum speed of 21,500 rpm. Another part of the biomass (2 L) was dried using thin-layer convective air drying (Mettmert UF260 model). A low temperature of 60°C was applied to prevent thermal degradation of biochemical compounds in the biomass according to Stramarkou et al. [35]. The paste was spread out in a 4 mm thick layer over 2 horizontal aluminum trays covered with parchment paper to prevent the material from sticking to the plate. In order to evaluate the performance of each drying technique, the moisture content and water activity (a_w) were measured using a moisture analyzer (Mettler Toledo HC103) and a water activity meter (Aqualab 4TE). Safe moisture levels ($< 8\%$) and water activity values (< 0.6) were reached in all dried *Nannochloropsis* biomasses [22]. Part of the dried biomass was stored in vacuum-sealed bags for analysis of free amino acids and 5'-nucleotides. Part of the dried biomass was diluted with mineral water (Cristaline) to 100 g DW/L and were frozen and stored at -80°C for volatile analysis and sensory evaluation (similar to intact and disrupted *Nannochloropsis* subsamples).

2.3. Volatile organic compounds analysis

Volatile organic compounds (VOCs) were determined using automated headspace solid-phase microextraction (HS-SPME) – gas chromatography–mass spectrometry (GC–MS) according to Coleman et al. [2]. For extraction, 12 mL of the *Nannochloropsis* paste (100 g DW/L) was hermetically sealed in 20 mL amber colored SPME vials to avoid photodegradation. Subsequently, the vial was incubated for 30 min at 40°C , followed by 30 min extraction at 40°C using a Supelco 50/30 μm DVB/CAR/PDMS fiber (Gerstel MPS sampler). The loaded fiber was desorbed in splitless mode (250°C , 3 min) and compounds were separated on a DB-5MS column ($30\text{ m} \times 250\text{ }\mu\text{m} \times 1\text{ }\mu\text{m}$; Agilent) using a helium flow rate of 1 mL/min. The oven temperature program was set as follows: start at 35°C , hold for 3 min, then raised to 220°C at a rate of 3.5°C/min (Agilent 7890A GC). The mass spectra in the electron impact ionization (EI) mode were generated at 70 eV and recorded in full scan mode ($35\text{--}250\text{ m/z}$) utilizing a 5975C inert XL mass spectrometer (Agilent).

To identify the aroma compounds, Unknown analysis of Masshunter (Agilent) was used. Identification of the aroma compounds was based on (1) spectral match, compared to the NIST (v2, 2011) and an in-house spectral database, and (2) retention index (RI), compared to the Aroma office 2D (v5.00.00, 2017, Gerstel) and an in-house RI database. RI calibration of the chromatogram was accomplished using an C8-C20 alkane standard solution (04070-1 mL, Sigma-Aldrich). Aroma compounds were identified if following criteria were met: (1) peak height is $>10^3$ (counts), (2) signal-to-noise ratio is higher than 3, (3) spectrum match probability is higher than 75 %, and (4) calculated experimental RI differs <15 from library RIs. Experimental Kovats RI, library RI and match factor information used for identification can be found in

Supplementary Information.

Semi-quantitative determination of the volatiles was performed by spiking 10 μL of the internal standard mix to each sample prior to SPME-GC-MS. An internal standard stock solution was prepared at 8.16 ng/ μL 2-methyl-3-heptanone (103128-5G, Sigma-Aldrich) and 87.5 ng/ μL methyl nonanoate (76368-1 mL, Sigma-Aldrich) in methanol (CL00.1377.2500, Chem-Lab Analytical). Similar to Isleten Hosoglu et al. [36], the area of the chromatographic peak of each identified volatile was divided by the area corresponding to the internal standard 2-methyl-3-heptanone. Areas corresponding to identified esters were divided by the area of internal standard methyl nonanoate. To calculate semi-quantitative concentrations of each volatile, the obtained responses were multiplied with the concentration of internal standard in the samples (6.8 $\mu\text{g/L}$ for 2-methyl-3-heptanone and 72.9 $\mu\text{g/L}$ for methyl nonanoate), assuming that all of the response factors were equal to one. Subsequently, these concentrations were divided by their individual odor threshold value (OTV) determined in water to calculate the odor activity values (OAVs) of each volatile. Since both concentration in the *Nannochloropsis* biomass and odor threshold of the VOC are taken into account, OAVs provide more insight on which volatile compounds contribute to the flavor. This approach for calculating semi-quantitative results is comparable to the approach described by Giri et al. [37] and Van Durme et al. [3]. In addition, all identified VOCs possess a specific odor descriptions which were retrieved from the The Good Scents Company database. The volatiles of the fresh intact and disrupted *Nannochloropsis* pastes were monitored for 24 h at ambient temperature in duplicate at five time points (0 h, 6 h, 12 h, 18 h and 24 h). Similar as in the sensory evaluation, the *Nannochloropsis* biomasses were defrosted from the -80°C for 1 h in the fridge and measured in triplicate.

2.4. Free amino acid analysis

Free amino acids (FAAs) were determined according to Coleman et al. [2]. Dried algae (0.2 g) is extracted with 8.5 mL UHPLC-MS water using a vortex for 15 s. Before extraction, 0.5 mL of the internal standard mixture was added containing 134.9 $\mu\text{g/mL}$ methionine-3,3,4,4- d_4 ; 34.1 $\mu\text{g/mL}$ *N*-methyl-L-valine; 146.3 $\mu\text{g/mL}$ histidine- $^{13}\text{C}_6^{15}\text{N}_3$; 282.7 $\mu\text{g/mL}$ L-glutamine- $^{13}\text{C}_5$; 1259.2 $\mu\text{g/mL}$ L-glutamic acid- $^{13}\text{C}_5$; 294.4 $\mu\text{g/mL}$ DL-Lysine- $^{13}\text{C}_{1,2}$; 435.7 $\mu\text{g/mL}$ L-aspartic acid- $^{15}\text{N}-2,3,3\text{-d}_3$; 634.1 $\mu\text{g/mL}$ L-asparagine- $^{15}\text{N}_2$; 316.8 $\mu\text{g/mL}$ L-alanine-2,3,3,3- d_4 and 430.9 $\mu\text{g/mL}$ homoarginine in UHPLC-MS water. FAAs were separated from proteins and peptides by precipitation using 1 mL 35 % sulfosalicylic acid (S2130-100G, Sigma-Aldrich). After 15 min of extraction at room temperature, samples were centrifuged for 10 min at 4000 g. Aliquots of 500 μL of the supernatants were diluted with 100 μL acidified (1%v/v formic acid) ammonium formate buffer (4 M) and 400 μL acetonitrile (ACN). Amino acids were separated according to van 't Land [38] by hydrophilic interaction chromatography (HILIC) using a LC-MS system. 3 μL of aqueous extract was injected into an Intrada HILIC column (100 mm $\times$ 3 mm $\times$ 3 μm) maintained at 37°C . A binary gradient at a flow rate of 0.6 mL/min was used, consisting of acidified (0.3%v/v formic acid) ACN and 80/20 (ACN ($\geq 99.95\%$)/ammonium formate ($\geq 99\%$)). Amino acids were detected using a Shimadzu triple quadrupole mass spectrometer (LCMS-8040) equipped with an electrospray ionization source (ESI). Nitrogen was used as nebulizer and argon as collision gas. Most amino acids were ionized in positive ESI and measured in multiple reaction monitoring (MRM), except for glycine in selected-ion monitoring (SIM). Aspartic acid was detected after negative ESI and SIM. Details on binary gradient, MRM and SIM can be found in Supplementary Information. The quantification was performed using calibration curves of 20 amino acids at 6 concentrations close to their taste threshold.

2.5. Free 5'-nucleotides analysis

Free 5'-nucleotides were extracted and analyzed according to

Moerdijk-Poortvliet et al. [39]. In brief, 50 mg dried algae and 5 mL Milli-Q were homogenized and extracted for 15 min at 35°C followed by centrifugation (3700 g, 20 min). The supernatant of the samples was supplied with 125 μL concentrated H_2SO_4 for acid precipitation followed by centrifugation (3700 g, 20 min). The supernatant was analyzed by means of High Performance Liquid Chromatography (HPLC) using a DIONEX Ultimate 3000 HPLC system equipped with a SIELC PrimeSep D mixed-mode column (150 mm $\times$ 4.6 mm $\times$ 5 μm) with a corresponding guard column (10 mm $\times$ 4.6 mm $\times$ 5 μm) and detected by Ultraviolet (UV) (DAD 3000). The injection volume was 10 μL and elution was performed isocratically at 0.8 mL/min, with 10 mM H_2SO_4 (pH 1.95) as the mobile phase and the detection wavelength was 260 nm. Quantification of 5'-nucleotides was achieved using a 7-point external calibration curve (5 to 1000 μM).

2.6. Equivalent umami concentration

The intensity of the umami taste of microalgae can be estimated by measuring the equivalent umami concentration (EUC) expressed in the amount of monosodium glutamate (MSG) per 100 g of product. The synergy effect between the umami amino acids and 5'-nucleotides is represented by the following equation [40]:

$$Y = \sum a_i b_i + 1218 (\sum a_i b_i) (\sum a_j b_j)$$

where Y is the EUC of the mixture expressed in g MSG/100 g; 1218 is a synergistic constant; a_i is the concentration (g/100 g) of each umami amino acid, aspartic acid (Asp) or glutamic acid (Glu); a_j is the concentration (g/100 g) of each umami 5'-nucleotide, inosine monophosphate (IMP), guanosine monophosphate (GMP) or adenosine monophosphate (AMP); b_i is the relative umami equivalent concentration (RUC) for each umami amino acid compared to MSG (1 for Glu and 0.077 for Asp) and b_j is the RUC for each umami 5'-nucleotide compared to IMP (1 for IMP; 2.3 for GMP and 0.18 for AMP).

2.7. Sensory evaluation

All sensory research performed in this study was in accordance with Ethical Standards of the Commission Flavor and Odor of ILVO (ECSG-ILVO). Prior to the taste sessions, samples were tested on microbial quality and safety (coagulase positive *S. aureus* enumerations, *E. coli* enumerations, *Salmonella* detection, *Listeria* spp. and *Listeria monocytogenes* detection) and trace element analysis (cadmium, lead, mercury, total arsenic, inorganic arsenic and iodine). Afterwards, a risk evaluation was made and evaluated by the ECSG-ILVO. Informed consent was obtained from all participants of the sensory evaluation. Because of safety concerns (possible bacterial growth and extensive lipid oxidation), the subsamples of intact and disrupted *Nannochloropsis* pastes that were stored for 24 h at ambient temperature could not be evaluated by the sensory panel.

The effect of drying and cell disruption on the flavor of *Nannochloropsis* was evaluated by a sensory panel consisting of 10 assessors (4 males, 6 females, aged between 24 and 39 years). The panelists were trained according to the ISO8586:2012 standard and had experience in evaluating algae (> 2 years). The taste lab is equipped according to the ISO standard 8589:2007 including 9 cubicles provided with touchscreens running Fizz (Fizz Biosystèmes) software, a table with faucet and sink and standardized LED lights, air-conditioning to standardize the temperature and ventilation of the lab. Prior to the sensory evaluation, the assessors were introduced to the different processed *Nannochloropsis* biomasses to identify sensory attributes. Afterwards, attributes were selected based on a panel discussion and sensory vocabulary found in the literature. The selected attributes are listed in Table 1. The intensity of the selected attributes were evaluated using a scale ranging from 0 (absent) to 10 (very strong). All five biomasses of *Nannochloropsis* (intact, disrupted, spray dried, oven dried and freeze-

Table 1

Sensory attributes associated to the *Nannochloropsis* biomasses, their descriptions and used reference products.

Sensory attributes	Description	Reference product
Grassy odor	The odor associated with sweet, freshly cut grass	0.2 g/L 1-hexen-3-ol solution
Musty odor	The odor associated with old hay, earthy-like	No reference
Sulfuric odor	The odor associated with onion, cabbage	0.001 g/L dimethyl sulfide solution
Fish oil odor	The odor associated with fish oil	Fish oil omega-3 (Ocean's Essentials)
Sour	The taste associated with citric acid	0.43 g/L citric acid
Bitter	The taste associated with caffeine	0.195 g/L caffeine
Umami	The taste associated with monosodium glutamate (MSG)	0.595 g/L MSG
Fishy taste	The taste of fish	10 g/L coalfish powder ^a

^a codfish powder (*Gadus morhua* meat) was obtained from Flandor Flavors International (Zulte, Belgium).

dried) were defrosted from -80°C for 1 h in the fridge (4°C) and diluted with mineral water (Cristaline, Jandun, France) to 10 g DW/L which is an acceptable concentration for the sensory evaluation of algae. Small amounts (20 mL) were presented in randomly coded and closed 30 mL screw-capped amber colored glass vials.

2.8. Statistical analysis

Using R (version 4.0.5), sensory scores fit the normal distribution based Quantile Quantile plot and histogram of the residuals. To determine significant differences between the *Nannochloropsis* biomasses for each sensory attribute, a linear mixed-effect model was conducted (lmer function, “lsmmeans” package). The effect of the assessors were considered random. To determine significant differences between the treatment and the sensory attributes, ANOVA (Type III) was used at an α risk of 5 %. This method is commonly applied to data from descriptive analysis [41]. Principal component analysis (PCA) was performed using MetaboAnalyst (version 5.0) to identify the similarities and differences in the volatile data (mean-centered and log transformed) and sensory data (mean-centered) of the different *Nannochloropsis* biomasses. The correlations between odor scores of the sensory evaluation and the VOCs detected by GC–MS were studied using partial least-squares regression (PLSR) (pls function, “pls” package). For the PLSR, the mean odor sensory scores (averaged over the panelists) and the mean chemical OAVs (average of 3 chemical replicates). Chemical OAVs were mean-centered and log transformed. To determine the number of components in the PLSR analysis, the leave-one-out cross-validation was used. The regression coefficients obtained from the PLSR-model were visualized in a heat map.

3. Results and discussion

3.1. Flavor profile of intact *Nannochloropsis* sp. biomass

The sensory evaluation indicated that the flavor of intact *Nannochloropsis* biomass was characterized by a sweet grassy odor with fishy odor notes and a weak umami taste (Fig. 3). This is comparable to the flavor description of freeze-dried *N. oceanica* obtained from Necton S.A. (Olhão, Portugal) dissolved in water (10 g DW/L) by Coleman et al. [2]. The flavor of *N. oceanica* distinguished itself from the other investigated microalgae because of its apparent sweet grassy odor. In an earlier study, Van Durme et al. [3] described the odor of *N. oculata* paste as grassy, cucumber-like and slightly fruity. These results might indicate that different *Nannochloropsis* strains have a similar flavor, however, this should be further investigated in a sensory evaluation performed by a trained panel.

A total of 82 compounds were detected in the VOCs analysis (Table 2). Alkanes (e.g. hexane) and alkenes (e.g. 1-hexene) were detected, but were excluded from the dataset because we assumed their contribution to the odor was negligible due to their high odor thresholds [42]. Partial least-squares regression (PLSR) was used to correlate the VOCs to the odor attributes obtained from the sensory analysis and the correlation coefficients were presented in a heat map (Fig. 2). Similar VOCs were linked to fish oil odor and sweet grassy odor attributes. These VOCs include fatty acid-derived unsaturated aldehydes (2-nonenal (E), 2,6-nonadienal (E,E), 2,4-heptadienal (E,E) and 2,4-decadienal (E,E)), unsaturated ketones (1-hexen-3-one and 1-octen-3-one), unsaturated alcohols (1-hexen-3-ol and 1-octen-3-ol) and the sulfur-containing compound dimethyl sulfide. These unsaturated aldehydes possess strong grassy, cucumber, fatty and fishy-like odors features, whereas the unsaturated ketones and alcohols have grassy and mushroom-like odor features. The odor of dimethyl sulfide is described as cabbage, onion and marine-like. The sweet grassy odor of *Nannochloropsis* was also correlated with esters (e.g. methyl 2-methylbutanoate) which have sweet and fruity odor features and saturated alcohols (e.g. 1-hexanol, 2-methylbutanol, 3-methylbutanol) which possess alcoholic, fruity and wine-like odor features. The musty odor of the *Nannochloropsis* biomass was highly correlated to the presence of alkyl aldehydes (e.g. 2-methylpropanal, 3-methylbutanal and 2-methylbutanal) which are characterized with musty, cocoa, coffee, nutty and malty-like odor features and furans (e.g. 2-ethylfuran and 2-ethyl-5-methylfuran) which possess with beany, cocoa, bread, coffee, nutty and malty-like odor features. In addition, the sulfur-containing compounds methanethiol and methional were also positively correlated with the musty odor attribute and are characterized with musty, earthy and decomposing odor properties.

The fatty acid-derived compounds (2,4-decadienal (E,E)), sulfur-containing compounds (dimethyl sulfide), saturated alcohols (3-methylbutanol) and esters (ethyl 2-methylpropanoate) had the highest OAVs and therefore strongly contribute to the characteristic odor of intact *Nannochloropsis*. Both fatty acid-derived and sulfur-containing VOCs are essential contributors of the aroma of fish and shellfish [7], which could explain the observed fishy-notes in the *Nannochloropsis* biomass. High amounts of fatty acid-derived compounds were also observed in *N. oceanica* and *N. oculata* in the studies of Coleman et al. [2] and Van Durme et al. [3], respectively. Similar to our study, the authors linked the grassy odor of the *Nannochloropsis* species to the presence of unsaturated aldehydes (e.g. 2,6-nonadienal (E,E) and 2-nonenal (E)) and unsaturated alcohols (1-hexen-3-ol and 1-octen-3-ol). These unsaturated fatty acid-derived volatiles can originate from the oxidation of PUFAs such as eicosapentaenoic acid (EPA) and linoleic acid (LA) in *Nannochloropsis* sp. [43].

The taste contribution of the FAAs in intact *Nannochloropsis* is mainly determined by the bitter/sweet-tasting proline (Pro), valine (Val) and lysine (Lys), sweet-tasting alanine (Ala) and umami-tasting glutamic acid (Glu). Full data on the FAAs can be found in Supplementary Information. Multiple chemical substances can elicit bitter and sweet taste, whereas umami taste receptors only respond to several chemical substances such as free amino acid Glu and 5'-nucleotides AMP. The Glu concentration measured in the intact *Nannochloropsis* is 1.35 mg Glu/g DW, which is comparable to the amount of the Glu found in *N. oceanica* (0.48 mg Glu/g DW) in the study of Coleman et al. [2]. However, the Glu concentration in *Nannochloropsis* is rather low in comparison to the other microalgae that have a more pronounced umami taste such as *Tetraselmis chuii* (5.67 mg Glu/g DW) and *Phaeodactylum tricornutum* (11.43 mg Glu/g DW). Derrien et al. [44] also observed higher free Glu concentration in five different microalgae (excluding *Nannochloropsis*), ranging from 2.0 to 13.5 mg Glu/g DW. Furthermore, low concentrations of the free 5'-nucleotide AMP (0.5 mg/g DW) were measured in intact *Nannochloropsis*, whereas free 5'-nucleotides GMP and IMP were not detected. Due to the low Glu and 5'-nucleotide concentrations in *Nannochloropsis*, low equivalent umami concentration (EUC) value were obtained (1.72 g MSG/100 g DW) which explains the weak umami taste

Table 2

Odor activity values (OAV) of all identified volatiles in the *Nannochloropsis* biomasses that were evaluated by the sensory panel. The mean OAVs are shown in the table ($n = 3$). The numbers of the volatile compound corresponds to those in Fig. 4. Library retention index, experimental Kovats retention index and spectral match factor information used for compound identification can be found in Supplementary Information. ND = not detected.

No	Compound	OTV (ppb)	Intact	Disrupted	Freeze-dried	Spray dried	Oven dried 60 °C
Sulfur-containing compounds							
X1	Methanethiol	0.02 ^c	192.6 ± 1.8	37 ± 2.2	72.1 ± 8.0	46.3 ± 2.3	68.5 ± 3.8
X2	Dimethyl sulfide	0.12 ^a	366.2 ± 25.2	138 ± 5.6	29.0 ± 1.9	ND	7.5 ± 0.5
X3	Dimethyl disulfide	0.29 ^a	10.8 ± 0.5	2.9 ± 2.9	7.9 ± 0.3	28.9 ± 0.9	2.7 ± 0.1
X4	Methional	0.2 ^b	ND	2.7 ± 0.2	ND	1.9 ± 0.0	12.3 ± 3.9
	Total		569.5 ± 25.2	180.7 ± 6.0	108.9 ± 8.3	77.1 ± 2.4	91.1 ± 3.9
Unsaturated aldehydes							
X5	2-Propenal, (E)-	3.6 ^a	3.7 ± 0.1	6.0 ± 0.2	3.0 ± 0.1	2.2 ± 0	5.1 ± 0.1
X6	2-Pentenal, (E)-	15 ^b	0.9 ± 0.1	1.3 ± 0.1	0.6 ± 0.1	0.7 ± 0	1.1 ± 0
X7	2-Hexenal, (E)-	17 ^d	0.9 ± 0	2.8 ± 0	1.9 ± 0.2	1.9 ± 0	4.4 ± 0.1
X8	4-Heptenal, (Z)-	0.8 ^b	0.9 ± 0	2.8 ± 0.1	4.1 ± 0.2	5.2 ± 0	9.1 ± 0
X9	2,4-Hexadienal, (E,E)-	10 ^b	ND	0.4 ± 0	0.2 ± 0	0.3 ± 0	0.8 ± 0
X10	2-Heptenal, (E)-	13 ^b	0.5 ± 0	1.3 ± 0	0.6 ± 0	0.7 ± 0	1.7 ± 0
X11	2,4-Heptadienal, (E,E)-	15.4 ^d	1.5 ± 0	7.9 ± 0	5.5 ± 0.4	3.6 ± 0	8.2 ± 0.1
X12	2-Octenal, (E)-	3 ^b	0.7 ± 0	1.8 ± 0	1.1 ± 0.1	1.0 ± 0	2.9 ± 0.1
X13	2,4-Octadienal, (E,E)-	15.4 ^d	ND	0.1 ± 0	0.1 ± 0	< 0.05	0.2 ± 0
X14	2,6-Nonadienal, (E,E)-	0.062 ^d	ND	10.0 ± 0.5	10.4 ± 0.5	7.7 ± 0.4	13.1 ± 1.0
X15	2-Nonenal, (E)-	0.08 ^b	ND	4.9 ± 0.1	ND	ND	11.4 ± 0.6
X16	2,4-Decadienal (E,E)-	0.07 ^b	32.7 ± 1.2	86.0 ± 6.9	61.5 ± 2.7	47.8 ± 2.4	72.4 ± 0.2
	Total		41.8 ± 1.2	125.3 ± 6.9	89.0 ± 2.8	71.1 ± 2.4	130.4 ± 1.1
Linear saturated aldehydes							
X17	Pentanal	0.4 ^a	21.6 ± 0.5	19.3 ± 0.2	17.9 ± 2.5	45.4 ± 0	33.2 ± 1.0
X18	Hexanal	2.4 ^e	22.6 ± 0.5	49.6 ± 0.5	33.9 ± 2.5	99.0 ± 0.6	139.0 ± 1.7
X19	Heptanal	0.8 ^b	5.2 ± 0.1	13.9 ± 0.4	11.5 ± 1.2	29.7 ± 0.9	53.6 ± 2.6
X20	Octanal	0.587 ^d	ND	1.1 ± 0.1	1.1 ± 0.1	1.9 ± 0.1	3.2 ± 0.1
X21	Nonanal	1 ^b	ND	ND	ND	ND	0.4 ± 0
	Total		49.5 ± 0.7	84.0 ± 0.7	64.4 ± 3.7	176.1 ± 1.1	229.3 ± 3.3
Unsaturated alcohols							
X22	1-Penten-3-ol	358.1 ^d	2.7 ± 0.1	3.5 ± 0	3.0 ± 0.3	2.1 ± 0	2.6 ± 0
X23	2-Penten-1-ol (E)-	400 ^b	0.4 ± 0	0.4 ± 0	0.5 ± 0	0.2 ± 0	0.2 ± 0
X24	2-Penten-1-ol (Z)-	400 ^b	1.3 ± 0	2.2 ± 0	2.1 ± 0.2	0.9 ± 0	0.9 ± 0
X25	1-Hexen-3-ol	3.9 ^e	6.3 ± 0.2	10.8 ± 0.2	7.8 ± 1.0	6.4 ± 0.1	6.8 ± 0.1
X26	1-Octen-3-ol	1 ^b	4.2 ± 0.2	16.8 ± 0.1	17.0 ± 1.6	9.7 ± 0.3	35.5 ± 0.3
	Total		14.9 ± 0.3	33.8 ± 0.3	30.4 ± 1.9	19.3 ± 0.3	46.0 ± 0.3
Unsaturated ketones							
X27	1-Penten-3-one	1 ^b	0.8 ± 0	1.5 ± 0	0.8 ± 0.1	1.1 ± 0	2.0 ± 0
X28	1-Hexen-3-one	1 ^b	7.6 ± 1.0	23.0 ± 0.3	6.7 ± 0.7	16.8 ± 0.9	46.2 ± 0.5
X29	1-Octen-3-one	0.9 ^b	0.9 ± 0.1	6.8 ± 0.4	2.3 ± 0.3	4.4 ± 0.2	29.0 ± 0.2
	Total		9.3 ± 1.0	31.3 ± 0.5	9.7 ± 0.7	22.3 ± 0.9	77.2 ± 0.6
Saturated alcohols							
X30	1-Propanol	31 ^a	0.1 ± 0	< 0.05	< 0.05	ND	ND
X31	3-Pentanol	5.5 ^a	14.3 ± 0.5	5.0 ± 0	2.9 ± 0.2	0.3 ± 0	0.6 ± 0
X32	3-Methylbutanol	1.69 ^a	60.5 ± 3.9	29.1 ± 0.5	22.3 ± 1.8	4.9 ± 0.2	3.7 ± 0.2
X33	2-Methylbutanol	1.69 ^a	13.7 ± 0.7	7.1 ± 0.1	5.3 ± 0.5	2.0 ± 0	0.5 ± 0
X34	1-Hexanol	24 ^a	4.2 ± 0.1	4.9 ± 0	4.2 ± 0.3	1.9 ± 0	0.6 ± 0
	Total		92.7 ± 4.0	46.2 ± 0.5	34.7 ± 1.9	9.1 ± 0.2	5.4 ± 0.2
Esters							
X35	Methyl ethanoate	170 ^a	< 0.05	< 0.05	< 0.05	< 0.05	ND
X36	Ethyl acetate	5 ^b	5.1 ± 0.2	2.6 ± 0	6.5 ± 0	ND	4.5 ± 0.1
X37	Ethyl propionate	10 ^e	0.3 ± 0	0.2 ± 0	1.2 ± 0.1	ND	0.4 ± 0
X38	Ethyl 2-methylpropanoate	0.1 ^b	ND	ND	7.9 ± 0.1	ND	3.3 ± 0.1
X39	Methyl 2-methylbutanoate	0.1 ^b	202.3 ± 14.1	69.9 ± 0.4	38.6 ± 3.4	4.8 ± 0.4	14.5 ± 0.6
X40	Ethyl 2-methylbutanoate	0.1 ^b	47.3 ± 1.9	20.7 ± 0.3	52.4 ± 2.3	3.1 ± 0	55.7 ± 0.4
X41	Ethyl 3-methylbutanoate	0.1 ^b	55.2 ± 1.4	23.9 ± 0.3	75.2 ± 4.4	3.8 ± 0.2	76.4 ± 0.2
X42	1-Butanol, 2-methyl-, acetate	0.75 ^a	1.4 ± 0	1.2 ± 0.1	0.7 ± 0	0.5 ± 0.1	0.8 ± 0
X43	Methyl hexanoate	70 ^b	< 0.05	< 0.05	0.1 ± 0	< 0.05	< 0.05
X44	Ethyl hexanoate	1 ^b	1.5 ± 0	1.9 ± 0	2.7 ± 0.1	0.2 ± 0	3.5 ± 0.1
X45	Methyl 2-ethylhexanoate	1 ^b	8.4 ± 0.1	6.5 ± 0.1	1.4 ± 0.1	1.9 ± 0.1	4.4 ± 0.1
X46	Methyl octanoate	200 ^b	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
X47	Ethyl octanoate	5 ^b	0.1 ± 0	0.4 ± 0	0.4 ± 0	< 0.05	1.1 ± 0
X48	Methyl decanoate	5 ^b	0.1 ± 0	0.2 ± 0	0.2 ± 0	0.1 ± 0	0.1 ± 0
X49	Methyl dodecanoate	5 ^b	0.1 ± 0	0.2 ± 0	0.2 ± 0	0.1 ± 0	0.1 ± 0
X50	Methyl tetradecanoate	5 ^b					

(continued on next page)

Table 2 (continued)

No	Compound	OTV (ppb)	Intact	Disrupted	Freeze-dried	Spray dried	Oven dried 60 °C
Total			321.9 ± 14.2	127.9 ± 0.6	187.4 ± 6.0	14.4 ± 0.4	165.0 ± 0.8
Alkyl aldehydes							
X51	2-Methylpropanal	0.1 ^b	4.0 ± 0.3	3.3 ± 0.1	5.9 ± 0.1	35.9 ± 0.2	74.4 ± 0.2
X52	3-Methylbutanal	1.1 ^d	7.0 ± 0.2	7.5 ± 0.1	7.2 ± 0.7	12.7 ± 0.5	41.2 ± 0
X53	2-Methylbutanal	0.2 ^e	10.5 ± 0.6	11.6 ± 0.4	16.0 ± 1.7	63.0 ± 2.7	182.4 ± 0.3
X54	3-Methyl-2-butenal	458.9 ^d	0.9 ± 0	0.6 ± 0	0.6 ± 0	0.9 ± 0.1	1.0 ± 0
Total			22.5 ± 0.7	23.1 ± 0.4	29.6 ± 1.9	112.5 ± 2.7	299.0 ± 0.4
Diketones							
X55	2,3-Butanedione	0.059 ^d	66.5 ± 1.3	10.2 ± 0.8	14.8 ± 0.3	16.4 ± 0.8	7.4 ± 0.7
X56	2,3-Octanedione	9.8 ^a	ND	0.3 ± 0	0.1 ± 0	0.1 ± 0	0.5 ± 0
Total			66.5 ± 1.3	10.5 ± 0.8	14.8 ± 0.3	16.5 ± 0.8	7.9 ± 0.7
Furans							
X57	2-Methylfuran	2.3 ^d	16.0 ± 0.9	13.4 ± 0.9	10.6 ± 0.5	15.5 ± 0.1	20.4 ± 1.8
X58	2-Ethylfuran	2.3 ^d	7.7 ± 0.2	8.0 ± 0.1	11.0 ± 0.6	7.8 ± 0.1	36.5 ± 0.6
X59	2-Ethyl-5-methylfuran	2.3 ^d	10.4 ± 0.2	3.8 ± 0.1	2.5 ± 0.3	1.4 ± 0	6.0 ± 0
X60	2-Butylfuran	5 ^d	0.5 ± 0	1.0 ± 0	1.7 ± 0.1	0.4 ± 0	0.4 ± 0
X61	2-Pentylfuran	5.8 ^d	1.6 ± 0.1	4.7 ± 0.1	2.3 ± 0.3	4.4 ± 0.1	17.0 ± 0
Total			36.2 ± 0.9	30.8 ± 0.9	28.1 ± 0.9	29.4 ± 0.2	80.4 ± 1.9
Saturated ketones							
X62	3-Pentanone	28 ^a	30.8 ± 0.5	6.4 ± 0.3	10.6 ± 0.7	2.9 ± 0.1	0.9 ± 0
X63	3-Hexanone	24 ^a	0.6 ± 0	0.1 ± 0	0.2 ± 0	0.1 ± 0	< 0.05
X64	2-Hexanone	24 ^a	< 0.05	ND	< 0.05	< 0.05	ND
X65	3-Heptanone	0.75 ^a	37.6 ± 0.2	15.8 ± 0.7	48.5 ± 4.6	8.2 ± 0.1	13.3 ± 0.5
X66	2-Heptanone	0.75 ^a	5.0 ± 0.3	3.3 ± 0.1	8.1 ± 0.6	6.7 ± 0	21.8 ± 0.1
X67	3-Octanone	21.4 ^d	0.5 ± 0	0.6 ± 0	1.1 ± 0	0.1 ± 0	0.3 ± 0
X68	2-Octanone	50 ^d	0.1 ± 0	< 0.05	0.3 ± 0	0.1 ± 0	0.1 ± 0
Total			74.6 ± 0.6	26.3 ± 0.7	69.0 ± 4.7	18.2 ± 0.1	36.7 ± 0.5
Benzaldehydes							
X69	Benzaldehyde	1.5 ^a	10.0 ± 0.9	24.9 ± 0.4	12.4 ± 1.5	16.4 ± 0.2	28.3 ± 0.8
X70	Benzeneacetaldehyde	4 ^b	1.1 ± 0	0.6 ± 0	1.0 ± 0.1	0.8 ± 0	5.9 ± 0.1
X71	Benzaldehyde, 2,4-dimethyl-	123.2 ^d	ND	ND	ND	ND	< 0.05
Total			11.1 ± 0.9	25.5 ± 0.4	13.4 ± 1.5	17.3 ± 0.2	34.2 ± 0.8
Carotenoids-derived compounds							
X72	5-Hepten-2-one, 6-methyl-	50 ^b	0.3 ± 0	0.3 ± 0	0.6 ± 0	0.1 ± 0	0.2 ± 0
X73	Cyclohexanone, 2,2,6-trimethyl-	100 ^b	< 0.05	0.1 ± 0	0.1 ± 0	0.1 ± 0	0.2 ± 0
X74	Isophorone	0.3 ^b	4.8 ± 0.3	4.7 ± 0.1	7.5 ± 0.1	12.9 ± 0.3	19.0 ± 0.1
X75	Acetophenone	65 ^b	0.1 ± 0	0.1 ± 0	0.1 ± 0	0.1 ± 0	0.1 ± 0
X76	Cyclohexanol, 2,6-dimethyl-	100 ^b	< 0.05	< 0.05	< 0.05	0.1 ± 0	< 0.05
X77	α-Cyclocitral	5 ^b	ND	ND	0.1 ± 0	0.1 ± 0	0.4 ± 0
X78	Safranal	5 ^b	< 0.05	0.1 ± 0	0.1 ± 0	0.2 ± 0	2.1 ± 0
X79	β-Cyclocitral	5 ^b	0.1 ± 0	0.4 ± 0	0.6 ± 0	1.0 ± 0	2.0 ± 0
X80	β-Homocyclocitral	5 ^b	ND	0.1 ± 0	0.1 ± 0	0.2 ± 0	0.4 ± 0
X81	β-Ionone	3.5 ^e	0.2 ± 0.1	1.2 ± 0.2	1.3 ± 0	1.9 ± 0.4	4.3 ± 0.4
Total			5.6 ± 0.5	6.9 ± 0.3	10.5 ± 1.6	16.7 ± 0.9	28.8 ± 0.7
X82	Pyrazine, 2-ethenyl-6-methyl-	1 ^b	ND	ND	ND	4.5 ± 0.0	0.7 ± 0.0

Odor detection threshold values (OTV) determined in water (in ppb).

^a Murnane et al. [45]. ^b Leffingwell and Leffingwell [46]. ^c The Good Scents Company [47]. ^d Giri et al. [37]. ^e Czerny et al. [48].

in the intact *Nannochloropsis*.

3.2. Effect of drying on the flavor profile of *Nannochloropsis* sp. biomass

Intact *Nannochloropsis* paste was dried by applying three commonly used drying methods: freeze-drying, spray drying and thin-layer oven drying at 60 °C. The dried powders were re-dissolved in water for the sensory evaluation and were compared with those of intact *Nannochloropsis* paste that had immediately been transferred to a – 80 °C freezer after harvesting. Both thermal drying methods spray drying and oven drying influenced the flavor profile of the *Nannochloropsis* biomass while freeze-drying had no significant effect (Fig. 3). Oven drying produced a strong musty odor and bitter taste of the *Nannochloropsis* biomass ($p < 0.05$). Spray drying reduced the odor intensity ($p < 0.05$),

mostly the grassy odor.

Thin-layer oven drying at 60 °C resulted in the formation of different VOCs that were not detected or present in low concentrations in the intact *Nannochloropsis* biomass. A strong increase in alkyl aldehydes (e.g. 2-methylpropanal, 3-methylbutanal and 2-methylbutanal), benzaldehydes (benzaldehyde and benzeneacetaldehyde) and sulfur-containing compound methional were observed. These VOCs are typical Strecker aldehydes that arise from Strecker degradation in which amino acids react with a dicarbonyl compound generated from the Maillard reaction [49,50]. Additionally, higher amounts of furans (e.g. 2-ethylfuran and 2-pentylfuran) were measured in the oven dried *Nannochloropsis* biomass which could arise from the thermal degradations reactions of carbohydrates, ascorbic acid, amino acids, carotenoids and PUFAs [49]. Oven drying induced also strong lipid oxidation, which is

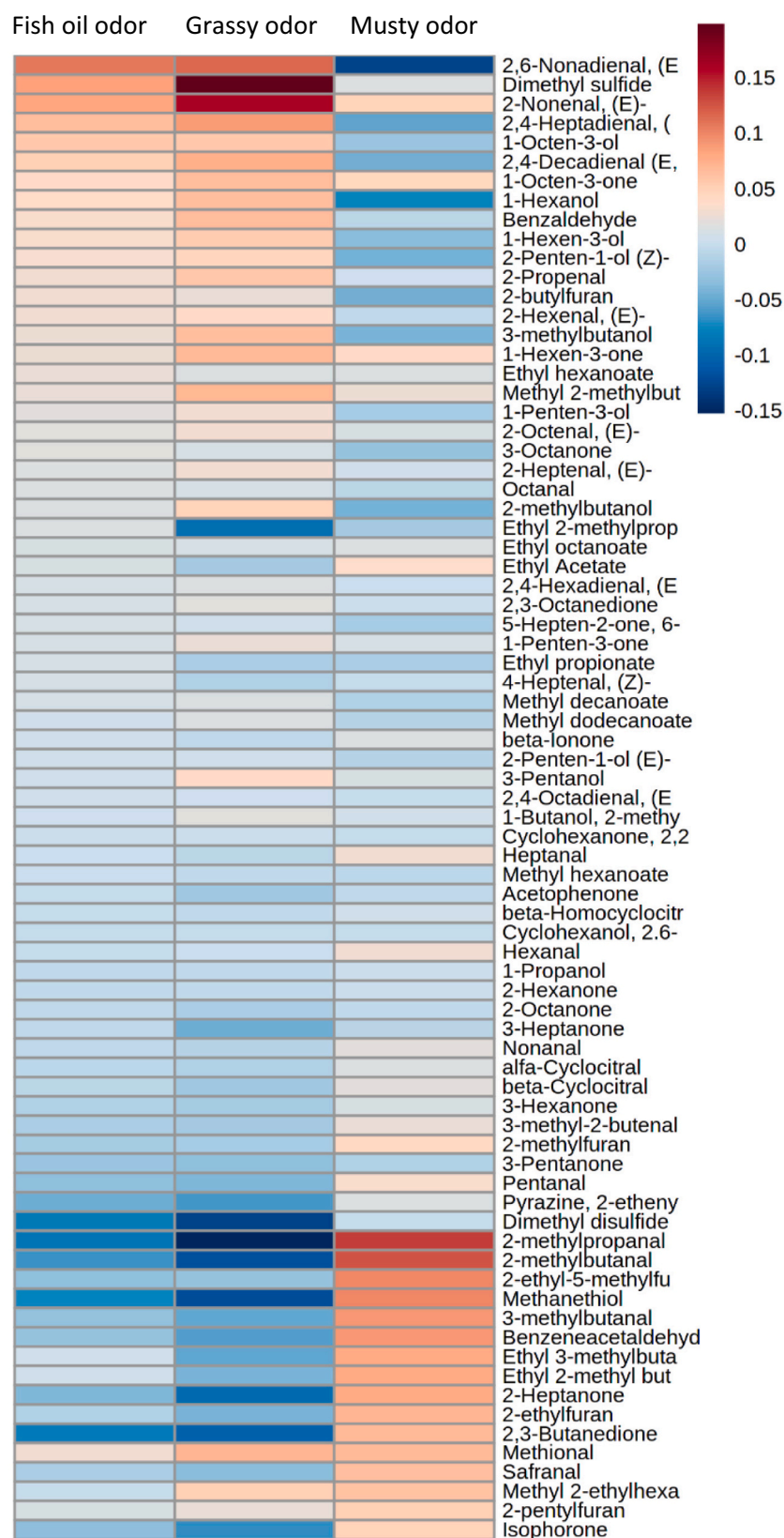


Fig. 2. Heat map of the correlation coefficient between each volatile and the odor attributes obtained by the PLSR analysis. High correlations are indicated in red and low correlations are indicated in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

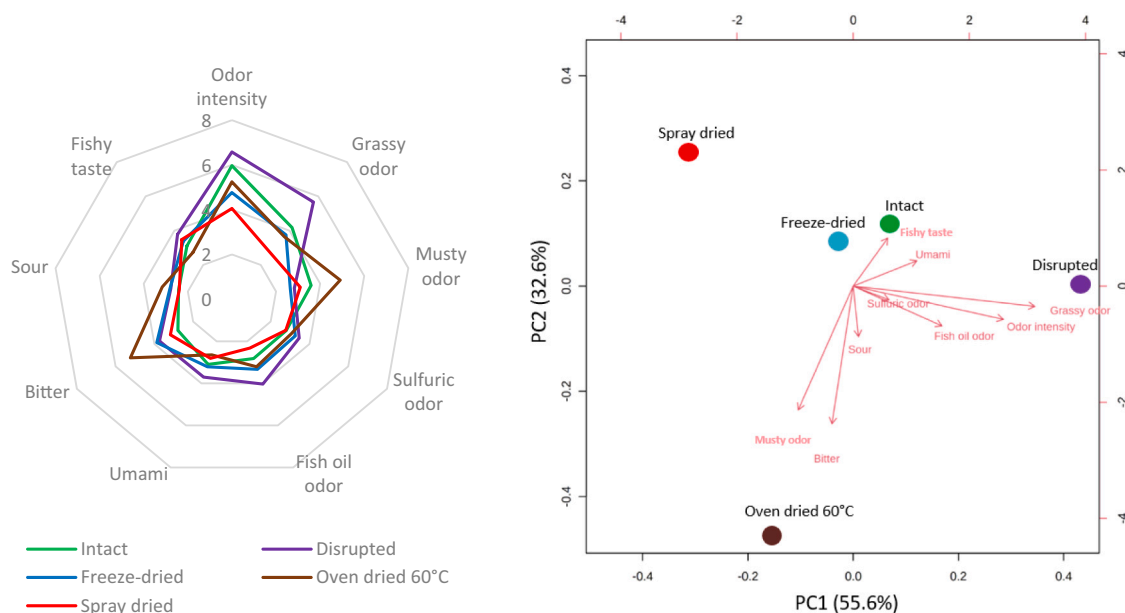


Fig. 3. The sensory evaluation of all *Nannochloropsis* biomasses visualized in a radar chart (left) and in a PCA plot (right). The PCA plot explains 88.2 % of the variance. Average scores over panelists ($n = 10$) were used.

related to the strong increase of fatty acid-derived compounds (e.g. 2-nonenal (E), 2,4-decadienal (E,E), 1-hexen-3-one 1-octen-3-one, 1-octen-3-ol, hexanal and heptanal). Finally, a rise in carotenoid-derived volatiles compounds (e.g. isophorone, safranal, β -cyclocitral and β -ionone) was observed in oven dried *Nannochloropsis*. Based on the PLSR results (Fig. 2), the strong musty odor in oven dried *Nannochloropsis* is mainly attributed to the high amounts of the Strecker aldehydes (e.g. 2-methylpropanal, 3-methylbutanal, 2-methylbutanal and benzeneacetaldehyde) and furans (e.g. 2-ethylfuran and 2-ethyl-5-methylfuran).

Spray drying resulted in a considerable decrease in the concentration of esters and saturated alcohols which are important for the sweet grassy odor of intact *Nannochloropsis* biomass. In addition, sulfur-containing compound dimethyl sulfide could not be detected in the spray dried biomass. This reduction in VOCs might explain the lower observed odor intensity of the spray dried biomass. Possibly, these VOCs are volatilized during the formation of the liquid droplets at the early atomization stage of spray drying at 140 °C [51]. An increase in Strecker aldehydes was also observed in the spray dried product, however, in a lower quantity compared to oven dried product. Remarkably, dimethyl disulfide and pyrazine, 2-ethenyl-6-methyl were detected during spray drying which can be formed at high temperatures (120–150 °C) [4,8]. However, their presence did not significantly influence the odor of the spray dried product.

Thin-layer oven drying at 60 °C created a hard dry surface on the *Nannochloropsis* biomass which makes the diffusion of moisture towards the upper surface difficult to evaporate [21]. Therefore, the drying of the microalgae paste in the oven takes a long time (approx. 7 h) which resulted in a more pronounced biomass degradation compared to the other drying methods [22]. In contrast to oven drying, the contact time between the *Nannochloropsis* paste and the hot air of 140 °C during spray drying is very short (approx. 35 s), causing only minor to moderate degradation in thermolabile components [52]. It was suggested that proteins are more resistant to high heat than to long heating times, which could explain the lower amount of Strecker aldehydes in spray dried *Nannochloropsis* compared to oven dried *Nannochloropsis* [53]. The slow drying rate in the oven promotes lipid oxidation in *Nannochloropsis*, which results in high amounts of fatty-acid derived volatiles. Possibly, the odor of the Strecker aldehydes and furans had a masking effect on these fatty-acid derived volatiles, as the odor of oven dried *Nannochloropsis* was characterized with a strong musty odor. Finally, exposure

to high temperature can destroy heat-sensitive carotenoids which could explain the highest amounts of carotenoid-derived volatiles detected in oven dried *Nannochloropsis* followed by spray dried *Nannochloropsis*.

The volatile composition of freeze-dried *Nannochloropsis* biomass was most similar to intact *Nannochloropsis*. Freeze-drying is generally believed to have a minimal impact on flavor profiles in comparison to most other food dehydration techniques [51]. However, some volatiles might be lost during the freeze-drying process due to the applied vacuum in the drying process [29]. Indeed, we observed a decrease in several VOCs compared to intact *Nannochloropsis*, including sulfur-containing compounds (e.g. methanethiol and dimethyl sulfide), esters (e.g. methyl 2-methylbutanoate), diketones (2,3-butanedione) and saturated alcohols (e.g. 3-methylbutanol and 2-methylbutanol). A principal component analysis (PCA) was carried out to visualize the effect of processing and storage on the volatile aroma composition of *Nannochloropsis*. The PCA plot in Fig. 4 shows that the first two principal components (PC) are explaining 68.8 % of the variance of the data, with component PC1 contributing 50.6 % and component PC2 18.2 %. From this PCA plot, it is clear that the thermal drying methods drastically changed the volatile profile of intact *Nannochloropsis* biomass, whereas freeze-dried *Nannochloropsis* is more similar to the intact biomass.

No significant differences were observed between the umami taste of the different dried *Nannochloropsis* and intact *Nannochloropsis* biomass. Low amount of Glu and Asp concentration were measured in the *Nannochloropsis* biomasses ranges from 0.59 to 2.18 mg Glu/g DW and 0.25 to 0.75 mg Asp/g DW, respectively. The free 5'-nucleotides IMP and GMP were not detected and only a small amount of AMP was present in the *Nannochloropsis* biomasses (0.51–1.08 mg AMP/g DW). As a result, low equivalent umami concentration (EUC) values were calculated (0.31–5.15 g MSG/100 g DW), which explains the weak umami taste in all *Nannochloropsis* biomasses. Probably, the differences EUC values between the different *Nannochloropsis* biomasses were too low to make a significant umami taste distinction. In the study of Latsos et al. [34], a difference in umami taste was observed between the *R. salina* biomasses cultivated at different cultivation pH ($p < 0.05$). Coleman et al. [2] observed significant difference in umami taste between several different microalgae species. In these studies, higher EUC values were calculated (> 8 g MSG/100 g DW) and the difference between the EUC values were larger (> 6 g MSG/100 g DW).

Importantly, a significantly stronger bitter taste was observed in the

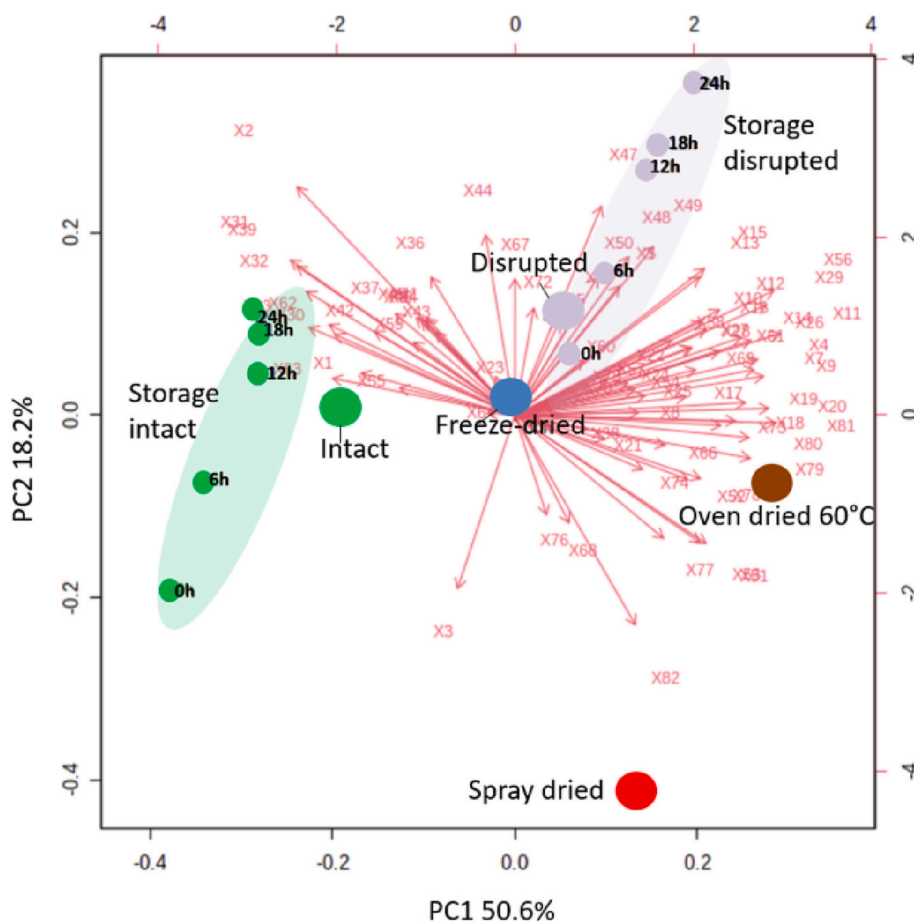


Fig. 4. PCA scores plot depicting distribution of identified volatile compounds in intact, disrupted, freeze-dried, oven dried and spray dried *Nannochloropsis* biomasses (large spheres). The smaller spheres visualize the fresh intact and disrupted *Nannochloropsis* pastes during the wet preservation at ambient temperature (0 h, 6 h, 12 h, 18 h and 24 h). The first two principal components (PC1 and PC2) explain 68.8 % of the variance. The different volatiles (X) are represented by a red arrow can be found in Table 2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

oven dried *Nannochloropsis* biomass which could not be explained by the bitter tasting FAAs. Possibly, the bitterness arise from extensive lipid oxidation which is more pronounced in oven dried *Nannochloropsis* (based on its volatile profile) compared to the other *Nannochloropsis* biomasses. Oxidation products of fatty acids (hydroperoxides) contributed to the bitterness of pea-protein isolates [54]. Furthermore, Refsgaard et al. [55] observed an increase in bitter taste of salmon during frozen storage caused by the oxidation of free PUFAs.

To conclude, the strong musty odor and bitter taste makes oven dried *Nannochloropsis* biomass less favorable for food purposes compared to intact, freeze-dried and spray dried *Nannochloropsis* biomass. Spray dried *Nannochloropsis* appears to possess the lowest odor intensity, as a result, the highest incorporation level might be reached using this biomass. Freeze drying retains most of the flavors of the intact

Nannochloropsis paste. Similar chemical reactions (e.g. Strecker degradation, Maillard reaction, thermal lipid oxidation, volatilization of VOCs) will take place during drying of other microalgae species, generating similar VOCs. However, it is expected that the VOC formation of the microalgae biomass will also depend on the biochemical composition such as fatty acid profile, amino acid composition, DMSP and carotenoid content of the microalgae species as these are important precursors for the generation of aroma during drying. Next to taste and aroma, color plays an important role in product development and consumer acceptance. Thermal drying resulted in a color change from intense green in freeze-dried *Nannochloropsis*, to bright yellowish green in spray dried *Nannochloropsis*, to dark brownish in oven dried *Nannochloropsis* (Fig. 5). This color difference is an indication of pigment (chlorophyll) deterioration and non-enzymatic browning produced by

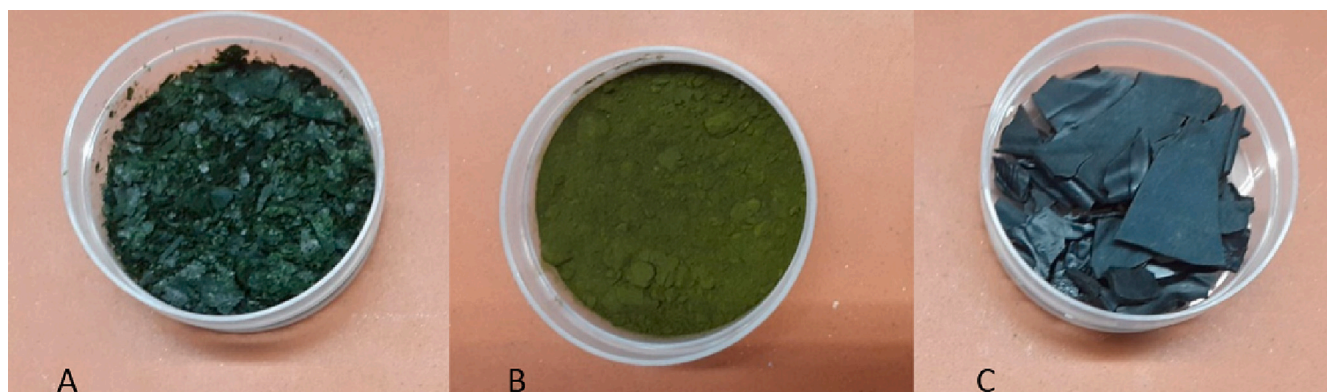


Fig. 5. Color changes of freeze-dried *Nannochloropsis* (A), spray dried *Nannochloropsis* (B) and thin-layer oven dried at 60 °C *Nannochloropsis* (C).

oxidized lipid/protein reactions during the heat treatment [56].

3.3. Effect of cell disruption and storage on the flavor profile of *Nannochloropsis* sp. biomass

Cell disruption using high pressure homogenization (HPH) has a major effect on the odor of intact *Nannochloropsis* biomass (Fig. 3). Significantly stronger grassy and fish oil odors were observed in the disrupted *Nannochloropsis* ($p < 0.05$). On the one hand, the volatile profile of disrupted *Nannochloropsis* is characterized with high amounts of unsaturated aldehydes (e.g. 2,4-decadienal (E,E), 2,6-nonadienal (E, E), 2-nonenal (E) and 2,4-heptadienal (E,E)), ketones (e.g. 1-hexen-3-one and 1-octen-3-one) and alcohols (e.g. 1-hexen-3-ol and 1-octen-3-ol). These unsaturated fatty acid-derived compounds, especially the unsaturated aldehydes, possess strong grassy, cucumber, fatty and fishy-like odors. On the other hand, similar amounts of Strecker aldehydes and furans were detected in the disrupted *Nannochloropsis* compared to intact *Nannochloropsis*.

Because the temperature during the disruption process was kept low ($< 30\text{ }^{\circ}\text{C}$), the unsaturated fatty acid-derived volatiles in disrupted *Nannochloropsis* were most likely formed by the increased enzymatic activity and autooxidation on the PUFAs rather than thermal lipid degradation. Gheysen et al. [57] reported significantly higher amount of free fatty acids in HPH-treated *Nannochloropsis* compared to freeze-dried *Nannochloropsis*. These free fatty acids, especially PUFAs, are very prone to lipoxygenase (LOX) and lyase activity, resulting in high amounts of fatty acid-derived VOCs [23]. The study of De Los Reyes et al. [58] strongly suggests that *N. gaditana* possess 15-LOX that can act on the EPA, resulting in 6-carbon volatiles formation such as 1-hexen-3-ol and 1-hexen-3-one. Most likely, other LOX were active in the disrupted *Nannochloropsis* biomass because of the detection of 8-carbon and 9-carbon volatiles such as 1-octen-3-ol, 1-octen-3-one and 2,6-nonadienal (E,E). The disruption of the cell wall/membrane may also result in a reduced oxidative stability due to the breakdown of carotenoids, which act as antioxidant in the microalgae [59]. The breakdown of carotenoids explains the small increase of β -cyclocitral and β -ionone in disrupted *Nannochloropsis*. Recently, Magpusao et al. [60] investigated the effect of HPH on the volatile profile of four different microalgae suspensions, including *Nannochloropsis* sp., testing different pressures (30, 60 and 90 MPa). In the study of Magpusao et al. [60], the harvested *Nannochloropsis* paste was freeze-dried and dissolved in water (80 g DW/L) followed by a single-pass through the HPH system. The HPH-treated pastes were frozen at $-20\text{ }^{\circ}\text{C}$ and thawed for VOC analysis, however,

no sensory analysis was performed. The strongest volatiles profile change was obtained in the *Nannochloropsis* treated at the highest pressure (90 MPa). Similar to our study, an increase in aldehydes (e.g. hexanal) was observed, however, the change in the volatile profile in the study of Magpusao et al. [60] was less drastic compared to our study. The difference might be explained by the HPH system settings in the study of Magpusao (90 MPa; 1 pass) compared to the HPH system settings applied in our study (100 MPa; 4 passes), which were chosen to completely disrupt the rigid cell wall of *Nannochloropsis* [16].

The aroma profile of both fresh intact and disrupted *Nannochloropsis* pastes change drastically during dark storage. Table 3 shows the total sum of the OAVs of the volatiles divided in specific groups similar to Table 2 to reveal the changes in aroma profiles of fresh intact and disrupted *Nannochloropsis* pastes stored at ambient temperature for 24 h. Complete data on all individual volatiles can be found in Supplementary Information. Notably, the formation rate of VOCs (expressed as a slope) strongly differs between the two microalgae pastes. During the dark storage of the fresh disrupted *Nannochloropsis* paste, the fatty acid-derived volatiles showed a rapid increase compared to fresh intact *Nannochloropsis* paste. The sum of the OAVs of all unsaturated aldehydes raised from 100.6 ± 1.5 to 343.1 ± 13.1 during the 24 h storage period (slope 9.68 with R^2 0.92) in the disrupted *Nannochloropsis*, while in intact cells the sum of the OAVs of all unsaturated aldehydes raises only from 4.4 ± 0.5 to 29.3 ± 7.8 after 24 h (slope 1.20 with R^2 0.70). Similar results were observed for the sum of the OAVs of all saturated aldehydes, unsaturated alcohols, ketones and furans. The strong increase of these fatty acid-derived volatiles indicates the importance of the cell wall/membrane integrity for the flavor of *Nannochloropsis* as the PUFAs can be oxidized by autooxidation and lipoxygenase/lyase activity more easy when the microalgae is ruptured. Cell rupture induces the activity of endogenous lipase which accelerates the formation of free fatty acids [17]. These free fatty acids act as a pro-oxidant which has an accelerating effect on the FAAs oxidation [61].

During the dark storage of the fresh intact *Nannochloropsis* paste, the saturated alcohols and esters showed a rapid increase compared to fresh disrupted *Nannochloropsis* paste. The sum of the OAVs of all saturated alcohols raises from 72.8 ± 11.3 to 145.6 ± 16.5 after 24 h (slope 3.09 with R^2 0.82) in intact *Nannochloropsis*, while in the disrupted *Nannochloropsis* OAVs increase from 21.4 ± 2.9 to 38.9 ± 2.9 after 24 h (slope 0.63 with R^2 0.56). Likewise, the sum of the OAVs of all esters raises from 276.6 ± 34.6 to 481.1 ± 31.1 after 24 h (slope 7.30 with R^2 0.77) in intact *Nannochloropsis*, while in the disrupted cells an increase from 59.9 ± 4.1 to 114.2 ± 11.6 is noted after 24 h (slope 2.00 with R^2 0.68).

Table 3
The change in flavor profile of the intact and disrupted *Nannochloropsis* pastes during dark storage at room temperature for 24 h. The total sum of the odor activity values (OAVs) of the volatiles divided in specific groups are shown. Complete data on all individual volatiles can be found in Supplementary Information. Slope and R^2 of the groups were calculated based on five time points (0 h, 6 h, 12 h, 18 h and 24 h) measured in duplicate in both intact and disrupted *Nannochloropsis* pastes. Free amino acids and 5'-nucleotides are measured in triplicate at beginning and end of the storage test and are expressed in mg/g DW.

Group totals	Intact <i>Nannochloropsis</i>				Disrupted <i>Nannochloropsis</i>			
	Time (0 h)	Time (24 h)	Slope	R^2	Time (0 h)	Time (24 h)	Slope	R^2
Sulfur-containing compounds	535.0 $\pm$ 164.7	713.4 $\pm$ 108.1	6.41	0.31	105.7 $\pm$ 23.8	329.1 $\pm$ 30.8	8.35	0.76
Unsaturated aldehydes	4.4 $\pm$ 0.5	29.3 $\pm$ 7.8	1.20	0.70	100.6 $\pm$ 1.5	343.1 $\pm$ 13.1	9.68	0.92
Linear saturated aldehydes	10.4 $\pm$ 0.7	24.2 $\pm$ 12.2	0.66	0.34	50.7 $\pm$ 3.7	272.1 $\pm$ 13.0	8.89	0.91
Unsaturated alcohols	3.2 $\pm$ 0.1	6.1 $\pm$ 1.2	0.12	0.76	19.6 $\pm$ 0.7	67.7 $\pm$ 1.4	1.96	0.97
Unsaturated ketones	3.5 $\pm$ 0.7	17.8 $\pm$ 7.6	0.64	0.79	21.2 $\pm$ 1.6	135.2 $\pm$ 10.9	4.65	0.91
Saturated ketones	8.2 $\pm$ 0.7	36.2 $\pm$ 9.8	1.33	0.64	12.8 $\pm$ 1.8	42.6 $\pm$ 0.9	1.26	0.93
Esters	276.6 $\pm$ 34.6	481.1 $\pm$ 31.1	7.30	0.77	59.9 $\pm$ 4.1	114.2 $\pm$ 11.6	2.00	0.68
Alkyl aldehydes	2.9 $\pm$ 0.7	9.2 $\pm$ 2.7	0.28	0.22	13.1 $\pm$ 1.1	21.8 $\pm$ 0.9	0.32	0.64
Diketones	13.9 $\pm$ 7.9	21.5 $\pm$ 0.8	0.15	0.02	5.6 $\pm$ 1.7	37.1 $\pm$ 0.5	1.29	0.91
Furans	12.7 $\pm$ 1.6	29.0 $\pm$ 2.0	0.68	0.64	15.5 $\pm$ 1.2	105.7 $\pm$ 1.9	3.83	0.98
Saturated ketones	8.2 $\pm$ 0.7	36.2 $\pm$ 9.8	1.33	0.64	12.8 $\pm$ 1.8	42.6 $\pm$ 0.9	1.26	0.93
Benzaldehydes	5.5 $\pm$ 0.2	6.4 $\pm$ 1.3	0.03	0.11	32.0 $\pm$ 3.9	44.1 $\pm$ 1.4	0.45	0.49
Carotenoids-derived compounds	9.0 $\pm$ 0.2	18.3 $\pm$ 0	0.13	0.79	5.2 $\pm$ 0.2	15.9 $\pm$ 1.8	0.42	0.94
Free amino acids (mg/g DW)	21.2 $\pm$ 0.1	17.2 $\pm$ 0.2			26.5 $\pm$ 0.1	42.7 $\pm$ 0.1		
Free 5'-nucleotides (mg/g DW)	0.5 $\pm$ 0.02	0.7 $\pm$ 0.02			0.6 $\pm$ 0.01	0.5 $\pm$ 0.01		
EUC (g MSG/100 g)	1.72	0.33			0.31	0.54		

Living eukaryotic *Nannochloropsis* cells present in the microalgae paste can maintain their energy demand through aerobic respiration depleting their intracellular polysaccharide reserves [62]. This oxygen consumption by the dense *Nannochloropsis* paste causes a fast drop of dissolved O₂ level in closed recipients [63]. Under anoxic conditions, saturated alcohols (e.g. 3-methylbutanol and 2-methylbutanol) and esters (e.g. methyl 2-methylbutanoate, ethyl 2-methylbutanoate and ethyl 3-methylbutanoate) can be formed via anaerobic respiration and fermentation reactions [64]. Similar to our study, an increase in saturated alcohols and esters was also observed during dark storage at 20 °C of wet *T-Isochrysis lutea* paste [14]. Furthermore, removal of oxygen has been proven to be an efficient method to retard lipid oxidation which is an additional explanation for the slower formation rate of fatty acid-derived VOC in intact *Nannochloropsis* compared to disrupted *Nannochloropsis* paste [65].

The sum of the OAVs of all sulfur-containing volatiles increased in both intact and disrupted *Nannochloropsis* pastes and is mainly attributed to dimethyl sulfide and methanethiol formation. In the intact *Nannochloropsis* paste, these sulfur-containing volatiles might be formed by microbial fermentation reactions on sulfur-containing compounds such as dimethylsulfoniopropionate (DMSP) released due to aging or grazing [64]. Methanethiol was also detected during wet preservation of *N. gaditana* paste [63] and found during storage of fresh seafood [66]. In the disrupted *Nannochloropsis* paste, the increase in sulfur-containing volatiles can be attributed to an increased DMSP lyase activity on released DMSP, resulting in dimethyl sulfide and methanethiol [67].

Based on the PLSR results presented in Fig. 2 and the change in volatile profile in Table 3, the odor of disrupted *Nannochloropsis* biomass would rapidly evolve to strong grassy, rancid, fatty and fish oil-like odors due to the strong increase of the fatty acid-derived volatiles. The increase of these fatty acid-derived volatiles is also commonly observed in fish rich in omega-3 PUFAs. Prost et al. [6] observed stronger fatty, fishy and rancid odor features in stored sardine (*Sardina pilchardus*) compared to fresh sardine which was attributed to an increase of fatty acid-derived volatiles such as 2,4-heptadienal-(E,E), 1-octen-3-ol, 2-heptenal (E), 4-heptenal (Z), 2-hexenal (E) and hexanal. In contrast, long stored intact *Nannochloropsis* biomass would evolve to overripe fruit, putrid and fermented-like odors, due to the rapid increase of esters, saturated alcohols and sulfur-containing volatiles and their corresponding odor descriptions. Sánchez-García et al. [68] reported that the odor of the fresh green seaweed *Ulva rigida* stored in the dark at 16 °C evolved from pleasant sea and marine to unwanted grassy and vegetable odor after 2 days. After 6 days, an unpleasant rotten odor was observed, whereas after 10 days of storage also fungi and moldy odors appeared. Similar to our study, the change in odor features of this green seaweed were attributed to increased (enzymatic) microbial activity.

Sensory evaluation showed no significant taste differences between intact and disrupted *Nannochloropsis*. Low concentrations of the free

amino acid Glu and the free 5'-nucleotide AMP were found between the intact and disrupted *Nannochloropsis* biomass, resulting in low EUC values (1.72 and 0.31 g MSG/100 g DW). During storage of intact *Nannochloropsis* paste, the FAAs content slightly decreased from 21.20 ± 0.14 mg/g DW to 17.2 ± 0.2 mg/g DW after 24 h, while in the disrupted *Nannochloropsis* the FAAs content increased from 26.54 ± 0.12 mg/g DW to 42.71 ± 0.13 mg/g DW after 24 h. Massoud et al. [27] indicated that the homogenization pressure could induce some conformational changes in protein structure of whey proteins and caseins, which increases their susceptibility to proteolysis, producing FAAs. The increase in total FAAs in disrupted *Nannochloropsis* paste would result in a stronger taste. Finally, a color change was observed from intense green (intact *Nannochloropsis*) to yellowish green because of cell disruption (Fig. 6).

Cell disruption of microalgae with a rigid cell wall/membrane such as *Nannochloropsis* is an important processing step to increase the digestibility and bioaccessibility of value compounds present in the microalgae such as PUFAs and carotenoids [69]. However, the increased grassy and fish oil odor intensities of the disrupted *Nannochloropsis* biomass could limit its incorporation level in food products. Storage of both intact and disrupted *Nannochloropsis* paste at ambient temperature should be avoided in order to maintain their palatability. This will also be the case for most other microalgae species. However, the severity of the impact of wet storage at ambient temperature on the flavor of microalgae might depend on factors including biochemical composition of the microalgae species (e.g. antioxidant capacity and fatty acid composition), the strength of the cell wall of the microalgae species and the bacterial load in the microalgae paste. Storage of the microalgae paste at lower temperatures (4 °C - 8 °C) could slowdown microbial activity, fermentation reactions and lipid oxidation [42,63], preventing unwanted VOC formation. Furthermore, the use of a sterilization step (e.g. pasteurization) to kill off the microorganism and to inhibit enzyme activity in the microalgae paste could reduce the formation of unwanted flavors during storage.

4. Conclusion

This study shows that thermal drying, cell disruption and storage strongly affects the flavor properties of *Nannochloropsis*. Slow thermal drying of *Nannochloropsis* paste such as thin-layer oven drying at 60 °C should be avoided to limit off-flavors such as bitterness and musty odor in the final food product. The strong musty odor is attributed to high amounts of Strecker aldehydes (e.g. 2-methylpropanal, 3-methylbutanal, 2-methylbutanal, benzeneacetaldehyde and methional) and furans. In contrast, spray drying reduced the odor intensity of the *Nannochloropsis* biomass which is explained by the loss of several volatile compounds (e.g. esters, saturated alcohols, dimethyl sulfide) during processing. Since microalgae have strong inherent flavoring capacity,



Fig. 6. Fresh intact *Nannochloropsis* paste (left) and fresh disrupted *Nannochloropsis* paste (right).

reducing their flavor intensity will enable higher incorporation levels in food which is favorable for nutritional purposes. Freeze-drying proved to be the best option to maintain the original flavor (sweet grassy aroma with fishy notes) of the fresh *Nannochloropsis* biomass.

Cell disruption is used to increase the digestibility and bioaccessibility of the microalgae biomass by disrupting the rigid cell wall. Disrupting the *Nannochloropsis* cells with the use of high pressure homogenization resulted in intense grassy and fish oil-like odor features. These odor features are attributed to high amounts of fatty acid-derived volatiles (e.g. 2,4-heptadienal (E,E), 2-nonenal (E), 2,6-nonadienal (E, E), 2,4-decadienal (E,E), 1-hexen-3-one, 1-octen-3-one, 1-octen-3-ol and 1-hexen-3-ol) and dimethyl sulfide. The strong odor intensity of disrupted *Nannochloropsis* could hinder high incorporation levels in food for nutritional purposes. The addition of disrupted *Nannochloropsis* biomass could be interesting as flavoring agent to improve the seafood flavor of food products which could help in the development of plant-based seafood alternatives. However, attention should be given when the disrupted *Nannochloropsis* is stored as its flavor properties could rapidly change to unwanted rancid and fatty flavors.

Storage of microalgae paste for food purposes should be limited due to the rapid generation of unwanted flavors which will reduce the palatability of the microalgae biomass. In large scale microalgae production plants, it is often impossible to reduce the storage time of the concentrated microalgae paste. Therefore, refrigerated storage of the microalgae paste is recommended to delay microbial deterioration and lipid oxidation until further processing. Furthermore, to avoid negative effects of these reactions on the microalgae flavor, other techniques (e.g. sterilization or pasteurization) should be further investigated. Although, it is expected that similar VOCs will be generated during drying and cell disruption of other microalgae species, the amount of the different individual VOCs produced from the microalgae might vary depending on factors including the biochemical composition of the microalgae biomass and the strength of the microalgae cell wall/membrane. Therefore, it is recommended to investigate the effect of processing on the flavor properties for other microalgae species. Next to processing and storage of microalgae, cultivation conditions could have an effect on the microalgae flavor. In this study, these effects were minimized by using a large culture of the same batch, however, the effect of cultivation conditions on the flavor of microalgae should be further investigated.

The results in this study provides necessary information for the optimal exploitation of *Nannochloropsis* biomass as a food ingredient or flavoring agent in order to develop tasteful food products. This will improve the consumer acceptance of the final product containing microalgae and thereby supporting the current movement of microalgae as natural and healthy food.

Statement of informed consent, human/animal rights

No conflicts or animal rights applicable. The sensory research procedures in this study followed the guidelines that constrain the use of human subjects from the Nuremberg code of ethics in medical research and the declaration of Helsinki (as revised in 2013).

Funding

This work was supported by the ValgOrize project, and is subsidized by the Interreg 2 seas programme 2014–2020 and co-financed by the European Regional Development Fund under subsidy contract Nr. 2S05-17.

CRediT authorship contribution statement

Bert Coleman: Conceptualization, Formal analysis, Investigation, Software, Methodology, Visualization, Data curation, Writing – original draft. **Christof Van Poucke:** Conceptualization, Investigation, Methodology, Data curation, Validation, Visualization, Software,

Supervision, Writing – review & editing. **Bavo De Witte:** Conceptualization, Investigation, Methodology, Data curation, Software, Supervision, Validation, Visualization, Writing – review & editing. **Valentina Casciaro:** Formal analysis, Methodology, Writing – review & editing. **Tanja Moerdijk-Poortvliet:** Formal analysis, Writing – review & editing. **Koenraad Muylaert:** Conceptualization, Investigation, Methodology, Supervision, Visualization, Writing – review & editing. **Johan Robbens:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors would like to thank the technicians of the Technology and Food Science Unit (ILVO) for their assistance in organizing the processing of the microalgae biomass in the Food Pilot. Thanks to Proviron Industries nv for cultivating and providing fresh *Nannochloropsis* paste. Thanks to the lab technicians of the marine analytical lab (MAL) of the Aquatic Environment and Quality department (ILVO) for their assistance in free amino acid analysis. Thanks to the lab technicians of the research group Marine Biobased Chemistry (HZ University) for their assistance in nucleotide analysis. Thanks to the group of Sciensano and the department of Microbiological Food Safety (ILVO) for performing the chemical and microbial safety tests. The authors would also like to thank Bart Ampe to help with statistical analyses.

References

- [1] Y. Torres-Tijji, F.J. Fields, S.P. Mayfield, Microalgae as a future food source, *Biotechnol. Adv.* 41 (August 2019) (2020), <https://doi.org/10.1016/j.biotechadv.2020.107536>.
- [2] B. Coleman, C. Van Poucke, B. Dewitte, A. Ruttens, T. Moerdijk-poortvliet, C. Latsos, J. Robbens, Potential of microalgae as flavoring agents for plant-based seafood alternatives, *Future Foods* 5 (January) (2022), 100139, <https://doi.org/10.1016/j.fufo.2022.100139>.
- [3] J. Van Durme, K. Goiris, A. De Winne, L. De Cooman, K. Muylaert, Evaluation of the volatile composition and sensory properties of five species of microalgae, *J. Agric. Food Chem.* 61 (46) (2013) 10881–10890, <https://doi.org/10.1021/jf403112k>.
- [4] S. Le Guen, C. Prost, M. Demaimay, Characterization of odorant compounds of mussels (*Mytilus edulis*) according to their origin using gas chromatography-olfactometry and gas chromatography-mass spectrometry, *J. Chromatogr. A* 896 (1–2) (2000) 361–371, [https://doi.org/10.1016/S0021-9673\(00\)00729-9](https://doi.org/10.1016/S0021-9673(00)00729-9).
- [5] F. Piveteau, S. Le Guen, G. Gandemer, J.P. Baud, C. Prost, M. Demaimay, Aroma of fresh oysters *Crassostrea gigas*: composition and aroma notes, *J. Agric. Food Chem.* 48 (10) (2000) 4851–4857, <https://doi.org/10.1021/jf991394k>.
- [6] C. Prost, A. Hallier, M. Cardinal, T. Serot, P. Courcoux, Effect of storage time on raw sardine (*Sardina pilchardus*) flavor and aroma quality, *J. Food Sci.* 69 (5) (2004), <https://doi.org/10.1111/j.1365-2621.2004.tb10732.x>.
- [7] F. Shahidi, J.R. Botta, Seafoods: chemistry, processing technology and quality, in: *Seafoods: Chemistry, Processing Technology and Quality*, 1994, <https://doi.org/10.1007/978-1-4615-2181-5>.
- [8] V. Varlet, X. Fernandez, Review. Sulfur-containing volatile compounds in seafood: occurrence, odorant properties and mechanisms of formation, *Food Sci. Technol. Int.* 16 (6) (2010) 463–503, <https://doi.org/10.1177/1082013210379688>.
- [9] D.W. Chen, M. Zhang, Non-volatile taste active compounds in the meat of chinese mitten crab (*Eriocheir sinensis*), *Food Chem.* 104 (3) (2007) 1200–1205, <https://doi.org/10.1016/j.foodchem.2007.01.042>.
- [10] R. Mendes, R. Quinta, M. Nunes, Changes in baseline nucleotides during ice storage of fish and crustaceans from the portuguese coast, *Eur. Food Res. Technol.* 212 (2001) 141–146, <https://doi.org/10.1007/s002170000222>.
- [11] J.L. Pérez-Lloréns, Microalgae: from staple foodstuff to avant-Garde cuisine, *Int. J. Gastronomy Food Sci.* 21 (April) (2020), 100221, <https://doi.org/10.1016/j.ijgfs.2020.100221>.
- [12] T. Lafarga, E. Mayre, G. Echeverria, I. Viñas, S. Villaró, F.G. Acien-fernández, I. Aguiló-aguayo, Potential of the microalgae *nannochloropsis* and *tetraselmis* for

- being used as innovative ingredients in baked goods, *LWT Food Sci. Technol.* 115 (July) (2019), 108439, <https://doi.org/10.1016/j.lwt.2019.108439>.
- [13] A. Skrede, L.T. Mydland, Ø. Ahlstrøm, K.I. Reitan, in: *Evaluation of Microalgae as Sources of Digestible Nutrients for Monogastric Animals* * 172580, 2011, pp. 131–142, <https://doi.org/10.22358/jafs/66164/2011>.
- [14] L. Balduyck, S. Bijttebier, C. Bruneel, G. Jacobs, S. Voorspoels, J. Van Durme, I. Foubert, Lipolysis in *T-Isochrysis lutea* during wet storage at different temperatures, *ALGAL* 18 (2016) 281–287, <https://doi.org/10.1016/j.algal.2016.07.003>.
- [15] J. Verspreet, L. Soetemans, Searching for appropriate storage conditions for short-term wet preservation of *Porphyridium purpureum*, *Appl. Sci.* (2020), <https://doi.org/10.3390/app10238315>.
- [16] T.M.M. Bernaerts, L. Gheysen, I. Foubert, M.E. Hendrickx, A.M. Van Loey, Evaluating microalgal cell disruption upon ultra high pressure homogenization, *Algal Res.* 42 (July) (2019), 101616, <https://doi.org/10.1016/j.algal.2019.101616>.
- [17] L. Balduyck, T. Stock, S. Bijttebier, C. Bruneel, G. Jacobs, S. Voorspoels, I. Foubert, Integrity of the microalgal cell plays a major role in the lipolytic stability during wet storage, *Algal Res.* 25 (2017) 516–524, <https://doi.org/10.1016/j.algal.2017.06.013>.
- [18] J. Velazquez-lucio, R.M. Rodríguez-jasso, L.M. Colla, A. Sáenz-galindo, C. N. Aguilar, D.E. Cervantes, H.A. Ruiz, in: *Microalgal Biomass Pretreatment for Bioethanol Production : A Review* 17, 2018, pp. 780–791, <https://doi.org/10.18331/BRJ2018.5.1.5>.
- [19] R. Halim, M.K. Danquah, P.A. Webley, Extraction of oil from microalgae for biodiesel production: a review, *Biotechnol. Adv.* 30 (3) (2012) 709–732, <https://doi.org/10.1016/j.biotechadv.2012.01.001>.
- [20] E. Ryckebosch, K. Muylaert, M. Eeckhout, T. Ruyssen, I. Foubert, Influence of drying and storage on lipid and carotenoid stability of the microalga *Phaeodactylum tricornutum*, *J. Agric. Food Chem.* 59 (20) (2011) 11063–11069, <https://doi.org/10.1021/jf2025456>.
- [21] E.G. Oliveira, J.H. Duarte, K. Moraes, V.T. Crexi, L.A.A. Pinto, Optimisation of *Spirulina platensis* convective drying: evaluation of phycocyanin loss and lipid oxidation, *Int. J. Food Sci. Technol.* 45 (8) (2010) 1572–1578, <https://doi.org/10.1111/j.1365-2621.2010.02299.x>.
- [22] H. Hosseinzad, S. Sokhansanj, C.J. Lim, Studying the drying mechanism of microalgae *Chlorella vulgaris* and the optimum drying temperature to preserve quality characteristics, *Dry. Technol.* 36 (9) (2018) 1049–1060, <https://doi.org/10.1080/07373937.2017.1369986>.
- [23] G. Pohnert, Phospholipase A2 activity triggers the wound-activated chemical defense in the diatom *Thalassiosira rotula*, *Plant Physiol.* 129 (1) (2002) 103–111, <https://doi.org/10.1104/pp.010974>.
- [24] S.B. Watson, Cyanobacterial and eukaryotic algal odour compounds: signals or by-products? A review of their biological activity, *Phycologia* 42 (4) (2003) 332–350, <https://doi.org/10.2216/i0031-8884-42-4-332.1>.
- [25] M.D. Keller, Dimethyl sulfide production and marine phytoplankton: the importance of species composition and cell size, *Biol. Oceanogr.* 5581 (1989) 375–382, <https://doi.org/10.1080/01965581.1988.10749540>.
- [26] D. Baranenko, V. Kolodyaznaya, Y. Broiyko, Effect of cold treatment on the amino acid composition of veal, *Agron. Res.* 12 (3) (2014) 705–716.
- [27] R. Massoud, S. Belgheisi, A. Massoud, Effect of high pressure homogenization on improving the quality of Milk and sensory properties of yogurt: a review, *Int. J. Chem. Eng. Appl.* 7 (2016) 66–70, <https://doi.org/10.7763/IJCEA.2016.V7.544>.
- [28] Z. Chen, H. Gao, W. Wu, H. Chen, Effects of fermentation with different microbial species on the umami taste of shiitake mushroom (*lentinus edodes*), *LWT* 141 (August 2020) (2021), 110889, <https://doi.org/10.1016/j.lwt.2021.110889>.
- [29] Y.H. Roos, CHAPTER 10 - food processing and storage, in: Y.H. Roos (Ed.), *Phase Transitions in Foods*, 1995, pp. 313–347, <https://doi.org/10.1016/B978-01259340-5/50010-9>.
- [30] M.J. Scholz, T.L. Weiss, R.E. Jinkerson, J. Jing, R. Roth, U. Goodenough, H. G. Gerken, in: *Ultrastructure and Composition of the Nannochloropsis gaditana* Cell 13(11), 2014, pp. 1450–1464, <https://doi.org/10.1128/EC.00183-14>.
- [31] S. Babuskin, K. Radhakrishnan, P. Azhagu Saravana Babu, M. Sukumar, M. A. Fayidh, K. Sabina, M. Sivarajan, Effects of Rosemary extracts on oxidative stability of chikkis fortified with microalgal biomass, *J. Food Sci. Technol.* 52 (6) (2015) 3784–3793, <https://doi.org/10.1007/s13197-014-1439-y>.
- [32] E. Rodríguez De Marco, M.E. Steffolani, M. Martínez, A.E. León, The use of nannochloropsis sp. As a source of omega-3 fatty acids in dry pasta: chemical, technological and sensory evaluation, *Int. J. Food Sci. Technol.* 53 (2) (2018) 499–507, <https://doi.org/10.1111/ijfs.13609>.
- [33] J. Fret, L. Roef, R. Blust, L. Diels, S. Tavernier, W. Vyverman, M. Michiels, Reuse of rejuvenated media during laboratory and pilot scale cultivation of nannochloropsis sp, *Algal Res.* 27 (June 2017) (2017) 265–273, <https://doi.org/10.1016/j.algal.2017.09.018>.
- [34] C. Latos, E. Wassenaar, T. Moerdijk, B. Coleman, J. Robbens, S. Van Roy, K. R. Timmermans, in: *Effect of pH on Rhodomonas salina Growth , Biochemical Composition , and Taste , Produced in Semi - large Scale Under Sunlight Conditions*, 2022, pp. 1215–1226, <https://doi.org/10.1007/s10811-022-02730-1>.
- [35] M. Stramarkou, S. Papadaki, K. Kyriakopoulou, M. Krokida, Effect of drying and extraction conditions on the recovery of bioactive compounds from *Chlorella vulgaris*, *J. Appl. Phycol.* 29 (6) (2017) 2947–2960, <https://doi.org/10.1007/s10811-017-1181-8>.
- [36] M. Isleten Hosoglu, Y. Karagül-Yuceer, O. Guner, Aroma characterization of heterotrophic microalgae *Cryptocodinium cohnii* using solid-phase microextraction and gas chromatography–mass spectrometry/olfactometry during different growth phases, *Algal Res.* 49 (April) (2020), 101928, <https://doi.org/10.1016/j.algal.2020.101928>.
- [37] A. Giri, K. Osako, T. Ohshima, Identification and characterisation of headspace volatiles of fish miso, a Japanese fish meat based fermented paste, with special emphasis on effect of fish species and meat washing, *Food Chem.* 120 (2) (2010) 621–631, <https://doi.org/10.1016/j.foodchem.2009.10.036>.
- [38] M. van 't Land, Fish Silage as Protein Ingredient in Animal Feeds : The Pioneering of Fishery Byproduct Utilisation in Belgium, Ghent University Faculty of Bioscience Engineering, 2019.
- [39] T.C.W. Moerdijk-Poortvliet, D.L.C. de Jong, R. Fremouw, S. de Reu, J.M. de Winter, K. Timmermans, G.C.H. Derksen, Extraction and analysis of free amino acids and 5'-nucleotides, the key contributors to the umami taste of seaweed, *Food Chem.* 370 (March 2021) (2022), 131352, <https://doi.org/10.1016/j.foodchem.2021.131352>.
- [40] S. Yamaguchi, T. Yoshikawa, S. Ikeda, T. Ninomiya, Measurement of the relative taste intensity of some L-α-amino acids and 5'-nucleotides, *J. Food Sci.* 36 (6) (1971) 846–849, <https://doi.org/10.1111/j.1365-2621.1971.tb15541.x>.
- [41] O.G. Mouritsen, L. Duelund, M.A. Petersen, A.L. Hartmann, M.B. Frøst, Umami taste, free amino acid composition, and volatile compounds of brown seaweeds, *J. Appl. Phycol.* 31 (2) (2019) 1213–1232, <https://doi.org/10.1007/s10811-018-1632-x>.
- [42] B. Matthäus, Oxidation of Edible Oils, 2010, <https://doi.org/10.1533/9780857090331.2.183>.
- [43] R. Huerlimann, R. de Nys, K. Heimann, Growth, lipid content, productivity, and fatty acid composition of tropical microalgae for scale-up production, *Biotechnol. Bioeng.* 107 (2) (2010) 245–257, <https://doi.org/10.1002/bit.22809>.
- [44] A. Derrien, L.J.M. Coiffard, C. Coiffard, Y. De Roeck-Holtzhauer, Free amino acid analysis of five microalgae, *J. Appl. Phycol.* 10 (2) (1998) 131–134, <https://doi.org/10.1023/A:1008003016458>.
- [45] Sharon S. Murnane, Alex H. Lohocky, P.D. Owens, Odor Thresholds for Chemicals with Established Occupational Health Standards, Retrieved from, 2nd Edition, American Industrial Hygiene Association (AIHA), 2013, <https://swesiaq.se/onetwebmedia/Dokument/ODORTHRESHOLDS.pdf>.
- [46] J.C. Leffingwell, D. Leffingwell, GRAS flavor chemicals-detection thresholds, *Perfumer Flavorist* 16 (1991) 1–19. Retrieved from, <http://www.leffingwell.com/odorthre.htm>.
- [47] The Good Scents Company, TGSC information system, Retrieved August 3, 2022, from 2021 website, <http://www.thegoodscentscompany.com/misc/about.html>, 2021.
- [48] M. Czerny, M. Christlbauer, M. Christlbauer, A. Fischer, M. Granvogl, M. Hammer, P. Schieberle, Re-investigation on odor thresholds of key food aroma compounds and development of an aroma language based on odor qualities of defined aqueous odorant solutions, *Eur. Food Res. Technol.* 228 (2008) 265–273, <https://doi.org/10.1007/s00217-008-0931-x>.
- [49] D.S. Mottram, The Maillard reaction: source of flavour in thermally processed foods, in: R.G. Berger (Ed.), *Flavours and Fragrances: Chemistry, Bioprocessing and Sustainability*, 2007, pp. 269–283, https://doi.org/10.1007/978-3-540-49339-6_12.
- [50] B.A. Smit, W.J.M. Engels, G. Smit, Branched chain aldehydes: production and breakdown pathways and relevance for flavour in foods, *Appl. Microbiol. Biotechnol.* 81 (6) (2009) 987–999, <https://doi.org/10.1007/s00253-008-1758-x>.
- [51] P. Fellows, Food processing technology, in: *Principles and Practice, Second Edition*, Woodhead Publishing Limited, 2000.
- [52] X.D. Chen, in: *Stickiness, Functionality, and Microstructure of Food Powders*, 2010, p. 3937, <https://doi.org/10.1080/07373930701397400>.
- [53] J.A. Nelson, DigitalCommons @ USU Postharvest Degradation of Microalgae : Effect of Temperature and Water Activity, 2015.
- [54] P. Gläser, V.K. Mittermeier-Kleßinger, A. Spaccasassi, T. Hofmann, C. Dawid, Quantification and bitter taste contribution of lipids and their oxidation products in pea-protein isolates (*Pisum sativum* L.), *J. Agric. Food Chem.* 69 (31) (2021) 8768–8776, <https://doi.org/10.1021/acs.jafc.1c02889>.
- [55] H.H.F. Refsgaard, P.M.B. Brockhoff, B. Jensen, Free polyunsaturated fatty acids cause taste deterioration of Salmon during frozen storage, *J. Agric. Food Chem.* 48 (8) (2000) 3280–3285, <https://doi.org/10.1021/jf000021c>.
- [56] R. Zamora, F. Hidalgo, The maillard reaction and lipid oxidation, *Lipid Technol.* 23 (2011) 59–62, <https://doi.org/10.1002/lite.201100094>.
- [57] L. Gheysen, N. Lagae, J. Devaere, K. Goiris, P. Goos, T. Bernaerts, I. Foubert, Impact of nannochloropsis sp. Dosage form on the oxidative stability of n-3 LC-PUFA enriched tomato purees, *Food Chem.* 279 (2018), <https://doi.org/10.1016/j.foodchem.2018.12.026>.
- [58] C. De Los Reyes, J. Ávila-Román, M.J. Ortega, A. De La Jara, S. García-Mauriño, V. Motilva, E. Zubía, Oxylipins from the microalgae *Chlamydomonas debaryana* and nannochloropsis gaditana and their activity as TNF-α inhibitors, *Phytochemistry* 102 (2014) 152–161, <https://doi.org/10.1016/j.phytochem.2014.03.011>.
- [59] T.L. Chacón-Lee, G.E. González-Mariño, Microalgae for “Healthy” foods: possibilities and challenges, *Compr. Rev. Food Sci. Food Saf.* 9 (6) (2010) 655–675, <https://doi.org/10.1111/j.1541-4337.2010.00132.x>.
- [60] J. Magpusao, I. Oey, B. Kebede, Changes in volatile and fatty acid compositions of selected microalgal suspensions with high pressure homogenization, *Algal Res.* 67 (August) (2022), 102867, <https://doi.org/10.1016/j.algal.2022.102867>.
- [61] D. Thurnham, Antioxidants in food: practical applications. Jan Pokorny, Nedyalka Yanishlieva and Michael Gordon (editors). 2001, Br. J. Nutr. (2002) 87, <https://doi.org/10.1079/BJN2001510>.
- [62] R. Halim, D.R.A. Hill, E. Hanssen, P.A. Webley, S. Blackburn, A.R. Grossman, G.J. O. Martin, Towards sustainable microalgal biomass processing: anaerobic

- induction of autolytic cell-wall self-ingestion in lipid-rich: nannochloropsis slurries, *Green Chem.* 21 (11) (2019) 2967–2982, <https://doi.org/10.1039/c8gc03186j>.
- [63] J. Verspreet, S. Kreps, L. Bastiaens, Evaluation of microbial load, formation of odorous metabolites and lipid stability during wet preservation of nannochloropsis gaditana concentrates, *Appl. Sci.* 10 (10) (2020), <https://doi.org/10.3390/app10103419>.
- [64] F. Jüttner, Physiology and biochemistry of odorous compounds from freshwater cyanobacteria and algae, *Water Sci. Technol.* 31 (11) (1995) 69–78, [https://doi.org/10.1016/0273-1223\(95\)00458-Y](https://doi.org/10.1016/0273-1223(95)00458-Y).
- [65] D.J. McClements, E.A. Decker, Lipid oxidation in oil-in-water emulsions: impact of molecular environment on chemical reactions in heterogeneous food systems, *J. Food Sci.* 65 (8) (2000) 1270–1282, <https://doi.org/10.1111/j.1365-2621.2000.tb10596.x>.
- [66] G. Duflos, V.M. Coin, M. Cornu, J.F. Antinelli, P. Malle, Determination of volatile compounds to characterize fish spoilage using headspace/mass spectrometry and solid-phase microextraction/gas chromatography/mass spectrometry, *J. Sci. Food Agric.* 86 (4) (2006) 600–611, <https://doi.org/10.1002/jsfa.2386>.
- [67] K.E. Achyuthan, J.C. Harper, R.P. Manginell, M.W. Moorman, Volatile metabolites emission by in vivo microalgae—an overlooked opportunity? *Metabolites* 7 (3) (2017) <https://doi.org/10.3390/metabo7030039>.
- [68] F. Sánchez-García, I. Hernández, V.M. Palacios, A.M. Roldán, Freshness quality and shelf life evaluation of the seaweed *Ulva rigida* through physical, chemical, microbiological, and sensory methods, *Foods* 10 (1) (2021), <https://doi.org/10.3390/foods10010181>.
- [69] T.M.M. Bernaerts, H. Verstreken, C. Dejonghe, L. Gheysen, I. Foubert, T. Grauwet, A.M. Van Loey, Cell disruption of nannochloropsis sp. improves in vitro bioaccessibility of carotenoids and ω 3-LC-PUFA, *J. Funct. Foods* 65 (2020), 103770, <https://doi.org/10.1016/j.jff.2019.103770>.